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Systems for generating geothermal power in an organic Rankine cycle operation during hydrocarbon production based on working fluid temperature

Abstract

Systems and methods for generating and a controller for controlling generation of geothermal power in an organic Rankine cycle (ORC) operation in the vicinity of a wellhead during hydrocarbon production to thereby supply electrical power to one or more of in-field operational equipment, a grid power structure, and an energy storage device. In an embodiment, during hydrocarbon production, a temperature of a flow of wellhead fluid from the wellhead or working fluid may be determined. If the temperature is above a vaporous phase change threshold of the working fluid, heat exchanger valves may be opened to divert flow of wellhead fluid to heat exchangers to facilitate heat transfer from the flow of wellhead fluid to working fluid through the heat exchangers, thereby to cause the working fluid to change from a liquid to vapor, the vapor to cause a generator to generate electrical power via rotation of an expander.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1982745	12/1933	Koenemann	N/A	N/A
3517208	12/1969	Williams et al.	N/A	N/A
3757516	12/1972	McCabe	N/A	N/A
3808794	12/1973	Wood	N/A	N/A
3875749	12/1974	Baciu	N/A	N/A
3908381	12/1974	Barber et al.	N/A	N/A
3988895	12/1975	Sheinbaum	N/A	N/A
4063417	12/1976	Shields	N/A	N/A
4079590	12/1977	Sheinbaum	N/A	N/A
4112687	12/1977	Dixon	N/A	N/A
4112745	12/1977	McCabe	N/A	N/A
4149385	12/1978	Sheinbaum	N/A	N/A
4157730	12/1978	Despois et al.	N/A	N/A
4191021	12/1979	Nakamura	N/A	N/A
4224796	12/1979	Stiel et al.	N/A	N/A
4228657	12/1979	Leo	N/A	N/A
4275563	12/1980	Kuroda	N/A	N/A
4292808	12/1980	Lohmiller	N/A	N/A
4356401	12/1981	Santi	N/A	N/A
4369373	12/1982	Wiseman	N/A	N/A
4484446	12/1983	Goldsberry	N/A	N/A

4542625	12/1984	Bronicki	N/A	N/A
4558568	12/1984	Hoshino et al.	N/A	N/A
4576005	12/1985	Force	N/A	N/A
4590384	12/1985	Bronicki	N/A	N/A
4982568	12/1990	Kalina	N/A	N/A
4996846	12/1990	Bronicki	N/A	N/A
5038567	12/1990	Mortiz	N/A	N/A
5117908	12/1991	Hofmann	N/A	N/A
5131231	12/1991	Trimble	N/A	N/A
5199507	12/1992	Westmoreland	N/A	N/A
5311741	12/1993	Blaize	N/A	N/A
5421157	12/1994	Rosenblatt	N/A	N/A
5440882	12/1994	Kalina	N/A	N/A
5483797	12/1995	Rigal et al.	N/A	N/A
5497624	12/1995	Amir et al.	N/A	N/A
5517822	12/1995	Haws et al.	N/A	N/A
5526646	12/1995	Bronicki et al.	N/A	N/A
5555731	12/1995	Rosenblatt	N/A	N/A
5570579	12/1995	Larjola	N/A	N/A
5595059	12/1996	Huber et al.	N/A	N/A
5598706	12/1996	Bronicki et al.	N/A	N/A
5660042	12/1996	Bronicki et al.	N/A	N/A
5661977	12/1996	Shnell	N/A	N/A
5671601	12/1996	Bronicki et al.	N/A	N/A
5685362	12/1996	Brown	N/A	N/A
5816048	12/1997	Bronicki et al.	N/A	N/A
5839282	12/1997	Bronicki et al.	N/A	N/A
5860279	12/1998	Bronicki et al.	N/A	N/A
RE36282	12/1998	Nitschke	N/A	N/A
5970714	12/1998	Bronicki et al.	N/A	N/A
5974804	12/1998	Sterling	N/A	N/A
6073448	12/1999	Lozada	N/A	N/A
6212890	12/2000	Amir	N/A	N/A
6536360	12/2002	O'Connor	N/A	N/A
6585047	12/2002	McClung	N/A	N/A
6691531	12/2003	Martinez	N/A	N/A
6695061	12/2003	Fripp et al.	N/A	N/A
6724687	12/2003	Stephenson et al.	N/A	N/A
6853798	12/2004	Weiss	N/A	N/A
6857268	12/2004	Stinger et al.	N/A	N/A
6857486	12/2004	Chitwood et al.	N/A	N/A
6989989	12/2005	Brasz et al.	N/A	N/A
7096665	12/2005	Stinger et al.	N/A	N/A
7174716	12/2006	Brasz et al.	N/A	N/A
7224080	12/2006	Smedstad	N/A	N/A
7225621	12/2006	Zimron et al.	N/A	N/A
7234314	12/2006	Wiggs	N/A	N/A
7237383	12/2006	Ahrens-Botzong et al.	N/A	N/A
7254949	12/2006	Brasz et al.	N/A	N/A
7281379	12/2006	Brasz	N/A	N/A

7287381	12/2006	Pierson et al.	N/A	N/A
7289325	12/2006	Brasz et al.	N/A	N/A
7313926	12/2007	Gurin	N/A	N/A
7320221	12/2007	Bronicki	N/A	N/A
7334410	12/2007	Creighton et al.	N/A	N/A
7337842	12/2007	Roddy et al.	N/A	N/A
7353653	12/2007	Bronicki	N/A	N/A
7428816	12/2007	Singh et al.	N/A	N/A
7472548	12/2008	Meksvanh	N/A	N/A
7493768	12/2008	Klaus et al.	N/A	N/A
7753122	12/2009	Curlett	N/A	N/A
7823386	12/2009	Zimron et al.	N/A	N/A
7891187	12/2010	Mohr	N/A	N/A
7891189	12/2010	Bottger et al.	N/A	N/A
7900450	12/2010	Gurin	N/A	N/A
7926276	12/2010	Dunn	N/A	N/A
7934383	12/2010	Gutierrez et al.	N/A	N/A
7942001	12/2010	Radcliff et al.	N/A	N/A
7950230	12/2010	Nishikawa et al.	N/A	N/A
7987676	12/2010	Ast et al.	N/A	N/A
8046999	12/2010	Doty	N/A	N/A
8096128	12/2011	Held et al.	N/A	N/A
8099198	12/2011	Gurin	N/A	N/A
8146360	12/2011	Myers et al.	N/A	N/A
8166761	12/2011	Moghtaderi et al.	N/A	N/A
8193659	12/2011	Bronicki et al.	N/A	N/A
8272217	12/2011	Lengert	N/A	N/A
8309498	12/2011	Funkhouser et al.	N/A	N/A
8371099	12/2012	Gutierrez et al.	N/A	N/A
8381523	12/2012	Zadok	N/A	N/A
8430166	12/2012	Danko	N/A	N/A
8438849	12/2012	Kaplan et al.	N/A	N/A
8459029	12/2012	Lehar	N/A	N/A
8511085	12/2012	Frey et al.	N/A	N/A
8528333	12/2012	Juchymenko	N/A	N/A
8534069	12/2012	Parrella	N/A	N/A
8555643	12/2012	Kalina	N/A	N/A
8555912	12/2012	Woolley et al.	N/A	N/A
8572970	12/2012	Matteson et al.	N/A	N/A
8578714	12/2012	Nagurny et al.	N/A	N/A
8596066	12/2012	Zimron et al.	N/A	N/A
8616000	12/2012	Parrella	N/A	N/A
8616001	12/2012	Held et al.	N/A	N/A
8616323	12/2012	Gurin	N/A	N/A
8656720	12/2013	Hardgrave	N/A	N/A
8667797	12/2013	Woodland	N/A	N/A
8667799	12/2013	Batscha	N/A	N/A
8674525	12/2013	Van den Bossche et al.	N/A	N/A
8680704	12/2013	Rooney	N/A	N/A
8707697	12/2013	Nitschke	N/A	N/A

8707698	12/2013	Conry	N/A	N/A
8708046	12/2013	Montgomery et al.	N/A	N/A
8720563	12/2013	Joseph et al.	N/A	N/A
8752382	12/2013	Lehar	N/A	N/A
8756908	12/2013	Sheridan et al.	N/A	N/A
8771603	12/2013	Harless et al.	N/A	N/A
8783034	12/2013	Held	N/A	N/A
8791054	12/2013	Deville	N/A	N/A
8820075	12/2013	Kaminsky	N/A	N/A
8820079	12/2013	Zyhowski et al.	N/A	N/A
8839857	12/2013	Schultz et al.	N/A	N/A
8841041	12/2013	Biederman et al.	N/A	N/A
8850814	12/2013	Kaplan et al.	N/A	N/A
8857186	12/2013	Held	N/A	N/A
8869531	12/2013	Held	N/A	N/A
8881805	12/2013	Klemencic	N/A	N/A
8919123	12/2013	Gibble et al.	N/A	N/A
8959914	12/2014	Kasuya et al.	N/A	N/A
8984883	12/2014	Riley	N/A	N/A
8984884	12/2014	Xu et al.	N/A	N/A
9003798	12/2014	Yanagi	N/A	N/A
9014791	12/2014	Held	N/A	N/A
9016063	12/2014	Gaia et al.	N/A	N/A
9062898	12/2014	Held	N/A	N/A
9077220	12/2014	Kyle et al.	N/A	N/A
9080789	12/2014	Hamstra et al.	N/A	N/A
9091278	12/2014	Vermeersch	N/A	N/A
9109398	12/2014	Harris et al.	N/A	N/A
9115603	12/2014	Leibowitz et al.	N/A	N/A
9115604	12/2014	Bronicki	N/A	N/A
9118226	12/2014	Kacludis et al.	N/A	N/A
9121259	12/2014	Bryant et al.	N/A	N/A
9150774	12/2014	Reddy et al.	N/A	N/A
9181930	12/2014	Klemencic	N/A	N/A
9217370	12/2014	Wang et al.	N/A	N/A
9234522	12/2015	Jonsson et al.	N/A	N/A
9243616	12/2015	Lee et al.	N/A	N/A
9297367	12/2015	Ramaswamy et al.	N/A	N/A
9316404	12/2015	Gurin	N/A	N/A
9322300	12/2015	Mirmobin et al.	N/A	N/A
9331547	12/2015	Bronicki	N/A	N/A
9341084	12/2015	Xie et al.	N/A	N/A
9341086	12/2015	Batscha et al.	N/A	N/A
9359919	12/2015	Berry	N/A	N/A
9376937	12/2015	Goswami et al.	N/A	N/A
9394764	12/2015	Favilli et al.	N/A	N/A
9394771	12/2015	Wiggs	N/A	N/A
9403102	12/2015	Wu et al.	N/A	N/A
9441504	12/2015	Held	N/A	N/A
9458738	12/2015	Held	N/A	N/A

9488160	12/2015	Fisher et al.	N/A	N/A
9499732	12/2015	Reddy et al.	N/A	N/A
9512348	12/2015	Reyes et al.	N/A	N/A
9512741	12/2015	Myogan et al.	N/A	N/A
9574551	12/2016	Parrella, Sr. et al.	N/A	N/A
9587161	12/2016	Fisk, Jr.	N/A	N/A
9587162	12/2016	Fisk, Jr.	N/A	N/A
9638065	12/2016	Vermeersch et al.	N/A	N/A
9649582	12/2016	Shnell	N/A	N/A
9671138	12/2016	Batscha et al.	N/A	N/A
9683463	12/2016	Juchymenko	N/A	N/A
9726157	12/2016	Sweatman et al.	N/A	N/A
9726441	12/2016	Reissner et al.	N/A	N/A
9732634	12/2016	Hikichi et al.	N/A	N/A
9745870	12/2016	Johnson et al.	N/A	N/A
9759096	12/2016	Vermeersch	N/A	N/A
9762460	12/2016	Pawlowski et al.	N/A	N/A
9777602	12/2016	Juchymenko	N/A	N/A
9784140	12/2016	Huntington et al.	N/A	N/A
9784248	12/2016	Batscha et al.	N/A	N/A
9797273	12/2016	Nishiguchi et al.	N/A	N/A
9803803	12/2016	Adams	N/A	N/A
9816402	12/2016	Kauffman et al.	N/A	N/A
9816443	12/2016	Sheridan et al.	N/A	N/A
9829194	12/2016	Aumann et al.	N/A	N/A
9840662	12/2016	Pascarella et al.	N/A	N/A
9845423	12/2016	Frantz et al.	N/A	N/A
9863282	12/2017	Hart et al.	N/A	N/A
9874112	12/2017	Giegel	N/A	N/A
9932861	12/2017	Preuss et al.	N/A	N/A
9932970	12/2017	Jeter	N/A	N/A
9957432	12/2017	Galindo et al.	N/A	N/A
9994751	12/2017	Hulse et al.	N/A	N/A
10005950	12/2017	Smith et al.	N/A	N/A
10024198	12/2017	Held et al.	N/A	N/A
10059870	12/2017	Joseph et al.	N/A	N/A
10060283	12/2017	Tomigashi et al.	N/A	N/A
10060302	12/2017	Weng et al.	N/A	N/A
10060652	12/2017	Tahara	N/A	N/A
10077683	12/2017	Close	N/A	N/A
10082030	12/2017	Genrup et al.	N/A	N/A
10113389	12/2017	Pandey et al.	N/A	N/A
10113535	12/2017	Conlon	N/A	N/A
10138405	12/2017	Kulkarni et al.	N/A	N/A
10138560	12/2017	Reyes et al.	N/A	N/A
10221770	12/2018	Sheridan	N/A	N/A
10227893	12/2018	McCune et al.	N/A	N/A
10234183	12/2018	Hashimoto	N/A	N/A
10247044	12/2018	Barmeier et al.	N/A	N/A
10247046	12/2018	Schuster et al.	N/A	N/A

10267184	12/2018	Bowan et al.	N/A	N/A
10323545	12/2018	Johnson	N/A	N/A
10352197	12/2018	Grill et al.	N/A	N/A
10357726	12/2018	Qin et al.	N/A	N/A
10400635	12/2018	Johnson et al.	N/A	N/A
10435604	12/2018	Kontomaris et al.	N/A	N/A
10436075	12/2018	Freund et al.	N/A	N/A
10458206	12/2018	Al-Dossary et al.	N/A	N/A
10465104	12/2018	Ravi et al.	N/A	N/A
10465491	12/2018	Moore	N/A	N/A
10472994	12/2018	Avadhanula et al.	N/A	N/A
10494897	12/2018	Pandey et al.	N/A	N/A
10495098	12/2018	Preuss et al.	N/A	N/A
10519814	12/2018	Quoilin	N/A	N/A
10527026	12/2019	Muir et al.	N/A	N/A
10563927	12/2019	Papadopoulos et al.	N/A	N/A
10570777	12/2019	Bowan	N/A	N/A
10570782	12/2019	Lintl et al.	N/A	N/A
10584660	12/2019	Sheridan et al.	N/A	N/A
10590324	12/2019	Kulkarni et al.	N/A	N/A
10590802	12/2019	McCune et al.	N/A	N/A
10598160	12/2019	Sumrall	N/A	N/A
10619520	12/2019	Juchymenko	N/A	N/A
10626709	12/2019	Al-Dossary	N/A	N/A
10670340	12/2019	Batscha et al.	N/A	N/A
10724805	12/2019	Barmeier et al.	N/A	N/A
10767904	12/2019	von Düring	N/A	N/A
10788267	12/2019	Dokic	N/A	N/A
10794292	12/2019	Kupratis et al.	N/A	N/A
10883388	12/2020	Held	N/A	N/A
10934895	12/2020	Held et al.	N/A	N/A
10947626	12/2020	Pinder et al.	N/A	N/A
10947839	12/2020	Cuthbert et al.	N/A	N/A
10975279	12/2020	Kontomaris et al.	N/A	N/A
11022070	12/2020	Aumann et al.	N/A	N/A
11137169	12/2020	Buscheck et al.	N/A	N/A
11168673	12/2020	Younes et al.	N/A	N/A
11174715	12/2020	Atisele	N/A	N/A
11187112	12/2020	Held	N/A	N/A
11187212	12/2020	Bodishbaugh et al.	N/A	N/A
11220932	12/2021	Kontomaris et al.	N/A	N/A
11236735	12/2021	Bodishbaugh et al.	N/A	N/A
11255315	12/2021	Bodishbaugh et al.	N/A	N/A
11255576	12/2021	Higgins et al.	N/A	N/A
11274660	12/2021	Radke	N/A	N/A
11274663	12/2021	Bodishbaugh et al.	N/A	N/A
11280322	12/2021	Bodishbaugh et al.	N/A	N/A
11293414	12/2021	Bodishbaugh et al.	N/A	N/A
11326479	12/2021	Radke	N/A	N/A
11326550	12/2021	Bodishbaugh et al.	N/A	N/A

11359576	12/2021	Bodishbaugh et al.	N/A	N/A
11359612	12/2021	Bodishbaugh et al.	N/A	N/A
11365652	12/2021	Gaia et al.	N/A	N/A
11396828	12/2021	Chase	N/A	N/A
11421625	12/2021	Bodishbaugh et al.	N/A	N/A
11421663	12/2021	Bodishbaugh et al.	N/A	N/A
11480074	12/2021	Bodishbaugh et al.	N/A	N/A
11486330	12/2021	Bodishbaugh et al.	N/A	N/A
11486370	12/2021	Bodishbaugh et al.	N/A	N/A
11493029	12/2021	Bodishbaugh et al.	N/A	N/A
11542888	12/2022	Bodishbaugh et al.	N/A	N/A
11549402	12/2022	Bodishbaugh et al.	N/A	N/A
11572849	12/2022	Bodishbaugh et al.	N/A	N/A
11578706	12/2022	Bodishbaugh et al.	N/A	N/A
11592009	12/2022	Bodishbaugh et al.	N/A	N/A
11598320	12/2022	Bodishbaugh et al.	N/A	N/A
11624355	12/2022	Bodishbaugh et al.	N/A	N/A
11644014	12/2022	Bodishbaugh et al.	N/A	N/A
11644015	12/2022	Bodishbaugh et al.	N/A	N/A
11668209	12/2022	Bodishbaugh et al.	N/A	N/A
11680541	12/2022	Bodishbaugh et al.	N/A	N/A
11732697	12/2022	Bodishbaugh et al.	N/A	N/A
11761353	12/2022	Bodishbaugh et al.	N/A	N/A
11761433	12/2022	Bodishbaugh et al.	N/A	N/A
11773805	12/2022	Bodishbaugh et al.	N/A	N/A
11879409	12/2023	Bodishbaugh et al.	N/A	N/A
11905934	12/2023	Bodishbaugh et al.	N/A	N/A
11933279	12/2023	Bodishbaugh et al.	N/A	N/A
11933280	12/2023	Bodishbaugh et al.	N/A	N/A
11946459	12/2023	Bodishbaugh et al.	N/A	N/A
11959466	12/2023	Bodishbaugh et al.	N/A	N/A
11971019	12/2023	Bodishbaugh et al.	N/A	N/A
12049875	12/2023	Bodishbaugh	N/A	N/A
12060867	12/2023	Bodishbaugh	N/A	N/A
12104553	12/2023	Bodishbaugh	N/A	N/A
12110878	12/2023	Bodishbaugh	N/A	N/A
12135016	12/2023	Bodishbaugh	N/A	N/A
12140124	12/2023	Bodishbaugh	N/A	N/A
12146475	12/2023	Bodishbaugh	N/A	N/A
12163485	12/2023	Bodishbaugh	N/A	N/A
12180861	12/2023	Bodishbaugh	N/A	N/A
12305624	12/2024	Bodishbaugh et al.	N/A	N/A
12312981	12/2024	Bodishbaugh et al.	N/A	N/A
2002/0178723	12/2001	Bronicki et al.	N/A	N/A
2003/0010652	12/2002	Hunt	N/A	N/A
2003/0029169	12/2002	Hanna et al.	N/A	N/A
2004/0237890	12/2003	Bour	N/A	N/A
2005/0034467	12/2004	Varney	N/A	N/A
2005/0109495	12/2004	Cheng et al.	N/A	N/A
2005/0247056	12/2004	Cogswell et al.	N/A	N/A

2005/0247059	12/2004	Cogswell et al.	N/A	N/A
2006/0026961	12/2005	Bronicki	N/A	N/A
2006/0130480	12/2005	Lovelace	N/A	N/A
2007/0025854	12/2006	Moore et al.	N/A	N/A
2008/0095611	12/2007	Storage	N/A	N/A
2008/0168772	12/2007	Radcliff et al.	N/A	N/A
2008/0217523	12/2007	O'Sullivan	N/A	N/A
2009/0071155	12/2008	Boyapati	N/A	N/A
2009/0211253	12/2008	Radcliff et al.	N/A	N/A
2009/0217664	12/2008	Rapp et al.	N/A	N/A
2009/0313999	12/2008	Hunter	N/A	N/A
2009/0320477	12/2008	Juchymenko	N/A	N/A
2010/0018207	12/2009	Juchymenko	N/A	N/A
2010/0034684	12/2009	Ast	N/A	N/A
2010/0045042	12/2009	Hinders et al.	N/A	N/A
2010/0071366	12/2009	Klemencic	N/A	N/A
2010/0071409	12/2009	Kaart	N/A	N/A
2010/0077752	12/2009	Papile	N/A	N/A
2010/0077792	12/2009	Gurin	N/A	N/A
2010/0187319	12/2009	Isom et al.	N/A	N/A
2010/0192573	12/2009	Hamilton	N/A	N/A
2010/0194111	12/2009	Van den Bossche et al.	N/A	N/A
2010/0218930	12/2009	Proeschel	N/A	N/A
2010/0300093	12/2009	Doty	N/A	N/A
2010/0319354	12/2009	Guidati et al.	N/A	N/A
2011/0000210	12/2010	Miles	N/A	N/A
2011/0000227	12/2010	Kamiya	N/A	N/A
2011/0030404	12/2010	Gurin et al.	N/A	N/A
2011/0041502	12/2010	Zimron et al.	N/A	N/A
2011/0041505	12/2010	Kasuya et al.	N/A	N/A
2011/0083620	12/2010	Yoon	N/A	N/A
2011/0100003	12/2010	McLeod et al.	N/A	N/A
2011/0126539	12/2010	Ramaswamy et al.	N/A	N/A
2011/0138809	12/2010	Ramaswamy et al.	N/A	N/A
2011/0175358	12/2010	Langson	N/A	N/A
2011/0272166	12/2010	Hunt	N/A	N/A
2011/0314818	12/2010	Breen et al.	N/A	N/A
2012/0001429	12/2011	Saar et al.	N/A	N/A
2012/0042650	12/2011	Ernst et al.	N/A	N/A
2012/0111004	12/2011	Conry	N/A	N/A
2012/0131918	12/2011	Held	N/A	N/A
2012/0145397	12/2011	Schultz et al.	N/A	N/A
2012/0174581	12/2011	Vaughan et al.	N/A	N/A
2012/0174622	12/2011	Granier	N/A	N/A
2012/0192560	12/2011	Ernst et al.	N/A	N/A
2012/0198844	12/2011	Kaminsky	N/A	N/A
2012/0261092	12/2011	Heath et al.	N/A	N/A
2012/0291433	12/2011	Meng et al.	N/A	N/A
2012/0292112	12/2011	Lakic	N/A	N/A
2012/0292909	12/2011	Erikson	N/A	N/A

2012/0315158	12/2011	Klaus	N/A	N/A
2013/0041068	12/2012	Reddy et al.	N/A	N/A
2013/0067910	12/2012	Ishiguro et al.	N/A	N/A
2013/0091843	12/2012	Zyhowski et al.	N/A	N/A
2013/0129496	12/2012	Eckert	N/A	N/A
2013/0139509	12/2012	Berti	N/A	N/A
2013/0168089	12/2012	Berg et al.	N/A	N/A
2013/0168964	12/2012	Xu et al.	N/A	N/A
2013/0186089	12/2012	Bruckner	N/A	N/A
2013/0217604	12/2012	Fisk, Jr.	N/A	N/A
2013/0227947	12/2012	Bronicki et al.	N/A	N/A
2013/0247569	12/2012	Suter	N/A	N/A
2013/0298568	12/2012	Pierson et al.	N/A	N/A
2013/0299123	12/2012	Matula	N/A	N/A
2013/0299170	12/2012	Joseph et al.	N/A	N/A
2013/0341010	12/2012	Nevison	N/A	N/A
2014/0011908	12/2013	Reddy et al.	N/A	N/A
2014/0026574	12/2013	Leibowitz et al.	N/A	N/A
2014/0033713	12/2013	Juchymenko	N/A	N/A
2014/0057810	12/2013	Fisk, Jr.	N/A	N/A
2014/0087978	12/2013	Deville	N/A	N/A
2014/0102098	12/2013	Bowan et al.	N/A	N/A
2014/0102103	12/2013	Yamamoto	N/A	N/A
2014/0123643	12/2013	Ming	N/A	N/A
2014/0130498	12/2013	Randolph	N/A	N/A
2014/0158429	12/2013	Kader et al.	N/A	N/A
2014/0178180	12/2013	Sheridan	N/A	N/A
2014/0206912	12/2013	Iglesias	N/A	N/A
2014/0224469	12/2013	Mirmobin et al.	N/A	N/A
2014/0296113	12/2013	Reyes et al.	N/A	N/A
2014/0305125	12/2013	Wang et al.	N/A	N/A
2014/0360702	12/2013	Cook	N/A	N/A
2014/0366540	12/2013	Zyhowski et al.	N/A	N/A
2015/0021924	12/2014	Parella	N/A	N/A
2015/0047351	12/2014	Ishikawa et al.	N/A	N/A
2015/0135708	12/2014	Lutz et al.	N/A	N/A
2015/0192038	12/2014	Sharp et al.	N/A	N/A
2015/0226500	12/2014	Reissner et al.	N/A	N/A
2015/0252653	12/2014	Shelton, Jr.	N/A	N/A
2015/0300327	12/2014	Sweatman et al.	N/A	N/A
2015/0330261	12/2014	Held	N/A	N/A
2015/0345341	12/2014	Kacludis et al.	N/A	N/A
2015/0345482	12/2014	Klitzing et al.	N/A	N/A
2015/0361831	12/2014	Myers	N/A	N/A
2016/0003108	12/2015	Held et al.	N/A	N/A
2016/0010512	12/2015	Close	N/A	N/A
2016/0017758	12/2015	Vermeersch et al.	N/A	N/A
2016/0017759	12/2015	Gayawal et al.	N/A	N/A
2016/0040557	12/2015	Vermeersch et al.	N/A	N/A
2016/0047540	12/2015	Aumann et al.	N/A	N/A

2016/0061055	12/2015	Bowan	N/A	N/A
2016/0076405	12/2015	Hashimoto et al.	N/A	N/A
2016/0084115	12/2015	Ludewig et al.	N/A	N/A
2016/0130985	12/2015	O'Connor et al.	N/A	N/A
2016/0160111	12/2015	Smith et al.	N/A	N/A
2016/0177887	12/2015	Fischer	N/A	N/A
2016/0201521	12/2015	Karthauser	N/A	N/A
2016/0222275	12/2015	Galindo et al.	N/A	N/A
2016/0257869	12/2015	Kulkarni et al.	N/A	N/A
2016/0312646	12/2015	Juano	N/A	N/A
2016/0340572	12/2015	Pascarella et al.	N/A	N/A
2016/0369408	12/2015	Reyes et al.	N/A	N/A
2017/0058181	12/2016	Frantz et al.	N/A	N/A
2017/0058722	12/2016	Noureldin et al.	N/A	N/A
2017/0130614	12/2016	Held et al.	N/A	N/A
2017/0145815	12/2016	Cuthbert et al.	N/A	N/A
2017/0159504	12/2016	Ostrom et al.	N/A	N/A
2017/0175582	12/2016	McCune et al.	N/A	N/A
2017/0175583	12/2016	McCune et al.	N/A	N/A
2017/0211829	12/2016	Slack	N/A	N/A
2017/0226402	12/2016	Patil et al.	N/A	N/A
2017/0233635	12/2016	Pandey et al.	N/A	N/A
2017/0240794	12/2016	Iverson et al.	N/A	N/A
2017/0254223	12/2016	Goethals et al.	N/A	N/A
2017/0254226	12/2016	Heber et al.	N/A	N/A
2017/0261268	12/2016	Barmeier et al.	N/A	N/A
2017/0276026	12/2016	Barmeier et al.	N/A	N/A
2017/0276435	12/2016	Papadopoulos et al.	N/A	N/A
2017/0284230	12/2016	Juchymenko	N/A	N/A
2017/0314420	12/2016	Bowan et al.	N/A	N/A
2017/0321104	12/2016	Ravi et al.	N/A	N/A
2017/0321107	12/2016	Joseph et al.	N/A	N/A
2017/0362963	12/2016	Hostler et al.	N/A	N/A
2017/0370251	12/2016	McGrail	N/A	N/A
2018/0094548	12/2017	Jeter	N/A	N/A
2018/0128131	12/2017	Zyhowski et al.	N/A	N/A
2018/0135468	12/2017	Kayahara	N/A	N/A
2018/0179960	12/2017	Apte et al.	N/A	N/A
2018/0224164	12/2017	Lakic	N/A	N/A
2018/0274524	12/2017	Moncarz et al.	N/A	N/A
2018/0313232	12/2017	Auciello	N/A	N/A
2018/0313340	12/2017	Spadacini et al.	N/A	N/A
2018/0328138	12/2017	Pandey et al.	N/A	N/A
2018/0340450	12/2017	Avadhanula et al.	N/A	N/A
2018/0355703	12/2017	Al-Dossary	N/A	N/A
2018/0356044	12/2017	Monti et al.	N/A	N/A
2019/0048759	12/2018	Noureldin et al.	N/A	N/A
2019/0055445	12/2018	Kulkarni et al.	N/A	N/A
2019/0128147	12/2018	Liu et al.	N/A	N/A
2019/0128567	12/2018	Redfern	N/A	N/A

2019/0390660	12/2018	McBay	N/A	N/A
2020/0011426	12/2019	Juchymenko	N/A	N/A
2020/0025032	12/2019	McCune et al.	N/A	N/A
2020/0041071	12/2019	Werlen	N/A	N/A
2020/0095899	12/2019	Merswolke et al.	N/A	N/A
2020/0200123	12/2019	Aumann et al.	N/A	N/A
2020/0200483	12/2019	Ahlbom	N/A	N/A
2020/0217304	12/2019	Sumrall	N/A	N/A
2020/0232342	12/2019	McCune et al.	N/A	N/A
2020/0248063	12/2019	Stone	N/A	N/A
2020/0292240	12/2019	Chase	N/A	N/A
2020/0308992	12/2019	Juchymenko	N/A	N/A
2020/0309101	12/2019	Muir et al.	N/A	N/A
2020/0354839	12/2019	Pinder et al.	N/A	N/A
2020/0386212	12/2019	Atisele	N/A	N/A
2020/0399524	12/2019	Pisklak et al.	N/A	N/A
2021/0017439	12/2020	Ramirez Angulo et al.	N/A	N/A
2021/0047963	12/2020	Conde	N/A	N/A
2021/0062682	12/2020	Higgins et al.	N/A	N/A
2021/0071063	12/2020	Stone	N/A	N/A
2021/0140684	12/2020	Younes et al.	N/A	N/A
2021/0172344	12/2020	Juchymenko	N/A	N/A
2021/0205738	12/2020	Blomqvist et al.	N/A	N/A
2021/0215440	12/2020	Morris	N/A	N/A
2021/0285693	12/2020	Gerson	N/A	N/A
2021/0372668	12/2020	Buscheck et al.	N/A	N/A
2021/0396116	12/2020	Yeung	N/A	N/A
2021/0404309	12/2020	Yeung	N/A	N/A
2022/0090521	12/2021	Kontomaris et al.	N/A	N/A
2022/0186984	12/2021	Gaia et al.	N/A	N/A
2022/0389844	12/2021	Zemlak	N/A	N/A
2024/0026838	12/2023	Bodishbaugh et al.	N/A	N/A
2024/0093660	12/2023	Bodishbaugh et al.	N/A	N/A
2024/0159224	12/2023	Bodishbaugh	N/A	N/A
2024/0159225	12/2023	Bodishbaugh	N/A	N/A
2024/0191703	12/2023	Bodishbaugh	N/A	N/A
2024/0209844	12/2023	Bodishbaugh	N/A	N/A
2024/0218772	12/2023	Bodishbaugh	N/A	N/A
2024/0254973	12/2023	Bodishbaugh	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2007204830	12/2006	AU	N/A
2009238733	12/2012	AU	N/A
2011336831	12/2015	AU	N/A
2012306439	12/2016	AU	N/A
2014225990	12/2017	AU	N/A
2692629	12/2008	CA	N/A
2698334	12/2008	CA	N/A
2676502	12/2010	CA	N/A

2679612	12/2017	CA	N/A
2676502	12/2017	CA	N/A
2952379	12/2018	CA	N/A
204661610	12/2014	CN	N/A
103174473	12/2014	CN	N/A
102812212	12/2015	CN	N/A
103174475	12/2015	CN	N/A
106517718	12/2016	CN	N/A
107246550	12/2016	CN	N/A
107387176	12/2016	CN	N/A
107461221	12/2016	CN	N/A
108302946	12/2017	CN	N/A
108457609	12/2017	CN	N/A
207761721	12/2017	CN	N/A
209457990	12/2018	CN	N/A
209704638	12/2018	CN	N/A
111837006	12/2019	CN	N/A
111911255	12/2019	CN	N/A
113137286	12/2020	CN	N/A
113266815	12/2020	CN	N/A
113983844	12/2021	CN	N/A
114370309	12/2021	CN	N/A
216922244	12/2021	CN	N/A
10337240	12/2004	DE	N/A
102011006066	12/2011	DE	N/A
102012214907	12/2012	DE	N/A
102012014443	12/2013	DE	N/A
102013009351	12/2013	DE	N/A
102018201172	12/2018	DE	N/A
0652368	12/1994	EP	N/A
1507069	12/2004	EP	N/A
2530255	12/2011	EP	N/A
2201666	12/2012	EP	N/A
1573173	12/2012	EP	N/A
1713877	12/2012	EP	N/A
1869293	12/2012	EP	N/A
2222939	12/2012	EP	N/A
1706667	12/2013	EP	N/A
2167872	12/2015	EP	N/A
2446122	12/2016	EP	N/A
2478201	12/2016	EP	N/A
3102796	12/2017	EP	N/A
3514339	12/2018	EP	N/A
2550436	12/2018	EP	N/A
3464836	12/2019	EP	N/A
3631173	12/2019	EP	N/A
2948649	12/2019	EP	N/A
3540331	12/2019	EP	N/A
2738872	12/1996	FR	N/A
2336943	12/2002	GB	N/A

247090	12/2002	IN	N/A
256000	12/2004	IN	N/A
202111000822	12/2020	IN	N/A
08192150	12/1995	JP	N/A
2001183030	12/2000	JP	N/A
2009127627	12/2008	JP	N/A
2010166805	12/2009	JP	N/A
2010249501	12/2009	JP	N/A
2010249502	12/2009	JP	N/A
2011064451	12/2010	JP	N/A
4668189	12/2010	JP	N/A
2011069370	12/2010	JP	N/A
2011106459	12/2010	JP	N/A
2011137449	12/2010	JP	N/A
2013151931	12/2012	JP	N/A
2013238228	12/2012	JP	N/A
2014016124	12/2013	JP	N/A
2014080975	12/2013	JP	N/A
2014109279	12/2013	JP	N/A
2015149885	12/2014	JP	N/A
2016006323	12/2015	JP	N/A
2016105687	12/2015	JP	N/A
2016188640	12/2015	JP	N/A
2021167601	12/2020	JP	N/A
101126833	12/2011	KR	N/A
20120067710	12/2011	KR	N/A
20130023578	12/2012	KR	N/A
1691908	12/2016	KR	N/A
2075550	12/2019	KR	N/A
2185002	12/2019	KR	N/A
581457	12/2010	NZ	N/A
2006142350	12/2007	RU	N/A
191467	12/2012	SG	N/A
191468	12/2012	SG	N/A
192327	12/2012	SG	N/A
1993/001397	12/1992	WO	N/A
1994028298	12/1993	WO	N/A
2005014981	12/2004	WO	N/A
2005019606	12/2004	WO	N/A
2005049975	12/2004	WO	N/A
2005100755	12/2004	WO	N/A
2006/014609	12/2005	WO	N/A
2006/027770	12/2005	WO	N/A
2006060253	12/2005	WO	N/A
2006/092786	12/2005	WO	N/A
2006138459	12/2005	WO	N/A
2007048999	12/2006	WO	N/A
20070079245	12/2006	WO	N/A
2007137373	12/2006	WO	N/A
2008052809	12/2007	WO	N/A

2008106774	12/2007	WO	N/A
2008125827	12/2007	WO	N/A
2009017471	12/2008	WO	N/A
2009017474	12/2008	WO	N/A
2009027302	12/2008	WO	N/A
2009030283	12/2008	WO	N/A
2009/058112	12/2008	WO	N/A
2009095127	12/2008	WO	N/A
2009/142608	12/2008	WO	N/A
2010021618	12/2009	WO	N/A
2010/039448	12/2009	WO	N/A
2010/065895	12/2009	WO	N/A
2009017473	12/2009	WO	N/A
2010106089	12/2009	WO	N/A
2010127932	12/2009	WO	N/A
2010/143046	12/2009	WO	N/A
2010/143049	12/2009	WO	N/A
2011012047	12/2010	WO	N/A
2011/066032	12/2010	WO	N/A
2011073469	12/2010	WO	N/A
2011/061601	12/2010	WO	N/A
2011/103560	12/2010	WO	N/A
2011093854	12/2010	WO	N/A
2011/137980	12/2010	WO	N/A
2012060510	12/2011	WO	N/A
2012/079694	12/2011	WO	N/A
2012074940	12/2011	WO	N/A
2012/112889	12/2011	WO	N/A
2012/142765	12/2011	WO	N/A
2012/151447	12/2011	WO	N/A
2013/014509	12/2012	WO	N/A
2013059695	12/2012	WO	N/A
2013/082575	12/2012	WO	N/A
2013103592	12/2012	WO	N/A
2013110375	12/2012	WO	N/A
2013115668	12/2012	WO	N/A
2013/136131	12/2012	WO	N/A
2014019755	12/2013	WO	N/A
2014042580	12/2013	WO	N/A
2014053292	12/2013	WO	N/A
2014059235	12/2013	WO	N/A
2014065977	12/2013	WO	N/A
2014/124061	12/2013	WO	N/A
2014/167795	12/2013	WO	N/A
2014154405	12/2013	WO	N/A
2014159520	12/2013	WO	N/A
2014159587	12/2013	WO	N/A
2014160257	12/2013	WO	N/A
2014164620	12/2013	WO	N/A
2014164620	12/2013	WO	N/A

2014165053	12/2013	WO	N/A
2014165053	12/2013	WO	N/A
2014165144	12/2013	WO	N/A
2014191157	12/2013	WO	N/A
2015/040279	12/2014	WO	N/A
2015034987	12/2014	WO	N/A
2015/059069	12/2014	WO	N/A
2015078829	12/2014	WO	N/A
2015/135796	12/2014	WO	N/A
2015131940	12/2014	WO	N/A
2015/158600	12/2014	WO	N/A
2015192005	12/2014	WO	N/A
2016039655	12/2015	WO	N/A
2016049712	12/2015	WO	N/A
2016050365	12/2015	WO	N/A
2016050366	12/2015	WO	N/A
2016050367	12/2015	WO	N/A
2016050368	12/2015	WO	N/A
2016050369	12/2015	WO	N/A
2016/069242	12/2015	WO	N/A
2013103631	12/2015	WO	N/A
2016073245	12/2015	WO	N/A
2016/087920	12/2015	WO	N/A
2016099975	12/2015	WO	N/A
2016147419	12/2015	WO	N/A
2016/196144	12/2015	WO	N/A
2016204287	12/2015	WO	N/A
2017041147	12/2016	WO	N/A
2017065683	12/2016	WO	N/A
2017123132	12/2016	WO	N/A
2017/146712	12/2016	WO	N/A
2017/147400	12/2016	WO	N/A
2017203447	12/2016	WO	N/A
2018/044690	12/2017	WO	N/A
2018/107279	12/2017	WO	N/A
2018106528	12/2017	WO	N/A
2018/210528	12/2017	WO	N/A
2018217969	12/2017	WO	N/A
2018227068	12/2017	WO	N/A
2019004910	12/2018	WO	N/A
2019060844	12/2018	WO	N/A
2019067618	12/2018	WO	N/A
2019086960	12/2018	WO	N/A
2019/157341	12/2018	WO	N/A
2019155240	12/2018	WO	N/A
2019/178447	12/2018	WO	N/A
2020152485	12/2019	WO	N/A
2020153896	12/2019	WO	N/A
2020/186044	12/2019	WO	N/A
2020201843	12/2019	WO	N/A

2020/229901	12/2019	WO	N/A
2020/097714	12/2019	WO	N/A
2020239067	12/2019	WO	N/A
2020239068	12/2019	WO	N/A
2020239069	12/2019	WO	N/A
2020251980	12/2019	WO	N/A
2021004882	12/2020	WO	N/A
2021013465	12/2020	WO	N/A
2021096696	12/2020	WO	N/A
2021107834	12/2020	WO	N/A
2022/061320	12/2021	WO	N/A

OTHER PUBLICATIONS

Richter, Alexander, GreenFire Energy and Mitsui Oil Exploration Co. are partnering on a closed-loop geothermal pilot project in Japan, Think GeoEnergy, Apr. 6, 2021. cited by applicant

Edwards, Alex, Dallas Innovates, Hunt Energy Network's New Venture Will Put 50 Batteries Across Texas, Giving ERCOT a Portfolio of Energy Generation, Apr. 1, 2021. cited by applicant

Guo, Boyun, Petroleum Engineering, A Computer-Assisted Approach, Dec. 21, 2006. cited by applicant

Li, Tailu et al., Cascade utilization of low temperature geothermal water in oilfield combined power generation, gathering heat tracing and oil recovery, Applied Thermal Engineering 40 (2012). cited by applicant

Sherven, Bob, Automation Maximizes performance for shale wells, Oil&Gas Journal, 2013. cited by applicant

Hu, Kaiyong et al., A case study of an ORC geothermal power demonstration system under partial load conditions in Huabei Oilfield, China, ScienceDirect, 2017. cited by applicant

Liu, Xiaolei et al., A systematic study of harnessing low-temperature geothermal energy from oil and gas reservoirs, Elsevier, ScienceDirect, Energy, 2017. cited by applicant

Wang, Kai, et al., A comprehensive review of geothermal energy extraction and utilization in oilfields, Elsevier, ScienceDirect, Journal of Petroleum Science and Engineering, 2017. cited by applicant

Cutright, Bruce L., The Transformation of Tight Shale Gas Reservoirs to Geothermal Energy Production, Bureau of Economic Geology University of Texas, Austin Texas, Jun. 14, 2011. cited by applicant

Khennich, Mohammed et al., Optimal Design of ORC Systems with a Low-Temperature Heat Source, Entropy 2012, 14, 370-389; doi:10.3390/e14020370. cited by applicant

Dambly, Benjamin W., et al., The Organic Rankine Cycle for Geothermal Power Generation, Geothermal Energy, 2007. cited by applicant

Obi, John Besong, State of art on ORC applications for waste heat recovery and micro-cogeneration for installations up to 100kWe, Elsevier, Energy Procedia 82 (2015) 994-1001. cited by applicant

Obafunmi, Jaiyejeje Sunday, Thermodynamic Analysis of Organic Rankine Cycles, Eastern Mediterranean University Jul. 2014, Gazimağusa, North Cyprus. cited by applicant

Dong, Bensi et al., Potential of low temperature organic Rankine cycle with zeotropic mixtures as working fluid, Elsevier, ScienceDirect, Energy Procedia 105 (2017) 1489-1494. cited by applicant

Iqbal, MdArbab et al., Trilateral Flash Cycle (TFC): a promising thermodynamic cycle for low grade heat to power generation, Elsevier, ScienceDirect, Energy Procedia 160 (2019) 208-214. cited by applicant

Bao, Junjiang et al., A review of working fluid and expander selections for organic Rankine cycle, Elsevier, ScienceDirect, Renewable and Sustainable Energy Reviews 24 (2013) 325-342. cited by

applicant

Ajmotokan, Habeeb A. et al., Trilateral Flash Cycle for Recovery of Power from a Finite Low-Grade Heat Source, Proceedings of the 24th European Symposium on Computer Aided Process Engineering—ESCAPE Jun. 15-18, 24, 2014, Budapest, Hungary. Copyright © 2014 Elsevier B.V. cited by applicant

Hung Tzu-Chen, et al., The Development and Application of a Small-Scale Organic Rankine Cycle for Waste Heat Recovery, IntechOpen, 2019. cited by applicant

Kong, Rithy et al., Thermodynamic performance analysis of a R245fa organic Rankine cycle (ORC) with different kinds of heat sources at evaporator, Elsevier, ScienceDirect, Case Studies in Thermal Engineering 13 (2019) 100385. cited by applicant

Lukawski, Maciej Z. et al., Impact of molecular structure of working fluids on performance of organic Rankine cycles (ORCs), Sustainable Energy Fuels, 2017, 1, 1098. cited by applicant

Saleh, Bahaa et al., Working fluids for low-temperature organic Rankine cycles, Elsevier, ScienceDirect, Energy 32 (2007) 1210-1221. cited by applicant

Brasz, Lars J. et al., Ranking of Working Fluids for Organic Rankine Cycle Applications, Purdue University, Purdue e-Pubs, (2004). International Refrigeration and Air Conditioning Conference. Paper 722. cited by applicant

Miller, Patrick C., Research uses landfill gas tech for Bakken flaring solution, The Bakken magazine, Sep. 16, 2015. cited by applicant

Electratherm, Inc., Power+ Generator 4400B & 4400B+, Nov. 24, 2020. cited by applicant

Electratherm, Inc., Heat To Power Generation Base Load Renewable Energy, Mar. 2020. cited by applicant

Electratherm, Inc., Power+ Generator, Nov. 25, 2020. cited by applicant

Electratherm, Inc., Generating Clean Power From Waste Heat, Nov. 2020. cited by applicant

Electratherm, Inc., Power+ Generator, May 19, 2020. cited by applicant

Sneary, Loy et al., Gulf Coast Green Energy, Flare Gas Reduction Trial Using an Organic Rankine Cycle Generator, Jan. 11, 2016. cited by applicant

Enertime, ORC for Industrial Waste Heat Recovery, Aug. 2017. cited by applicant

Enogia, Generate power from your waste heat thanks to our ORC, 2019. cited by applicant

UTC Power, PureCycle, 200 Heat-to-Electricity Power System, 2004. cited by applicant

Rank, MT3 machine, Dec. 17, 2018. cited by applicant

Heat Recovery Solutions, Clean Cycle Containerized Solution, 2009. cited by applicant

Triogen BV, Specification: E-Box Engine Application, Feb. 15, 2019. cited by applicant

“From Waste Heat To High Performance”, PBOG (Permian Basin Oil and Gas Magazine), Apr. 26, 2013. cited by applicant

“Turning Waste Heat Into Clean Power; GNP's Expander System”, Great Northern Power Corporation, 2022. cited by applicant

Ng et al., “Thermo-Economic Performance of an Organic Rankine Cycle System Recovering Waste Heat Onboard an Offshore Service Vessel”, Journal of Marine Science and Engineering, May 14, 2020. cited by applicant

“First Flare Elimination Demonstration”, ElectraTherm, 2022. cited by applicant

International Search Report and Written Opinion for PCT/US2022/071472, May 9, 2022. cited by applicant

International Search Report and Written Opinion for PCT/US2022/071475, May 17, 2022. cited by applicant

International Search Report and Written Opinion for PCT/US2022/071474, Jun. 10, 2022. cited by applicant

International Search Report and Written Opinion for PCT/US2022/071486, Jun. 14, 2022. cited by applicant

International Search Report and Written Opinion for PCT/US2022/071323, Jun. 28, 2022. cited by

applicant

International Search Report and Written Opinion for PCT/US2022/071325, Jun. 28, 2022. cited by applicant

Invitation to Pay Additional Fees and Communication Relating to Results of Partial International Search for PCT/US2022/071329, mailed Jun. 27, 2022. cited by applicant

Invitation to Pay Additional Fees and Communication Relating to Results of Partial International Search for PCT/US2022/071327, mailed Jul. 4, 2022. cited by applicant

International Search Report and Written Opinion for PCT/US2022/071313, Jul. 5, 2022. cited by applicant

International Search Report and Written Opinion for PCT/US2022/071517, Jun. 27, 2022. cited by applicant

International Search Report and Written Opinion for PCT/US2022/071484, Jun. 27, 2022. cited by applicant

International Search Report and Written Opinion for PCT/US2022/071319, Jul. 12, 2022. cited by applicant

International Search Report and Written Opinion for PCT/US2022/071480, Aug. 3, 2022. cited by applicant

International Search Report and Written Opinion for PCT/US2022/071482, Aug. 2, 2022. cited by applicant

Li et al., “Thermodynamic analysis of subcritical/transcritical ORCs with metal-organic heat carriers for efficient power generation from low-grade thermal energy”, Energy 255 (2022) 124519. cited by applicant

International Search Report and Written Opinion for PCT/US2022/071327, Aug. 29, 2022. cited by applicant

International Search Report and Written Opinion for PCT/US2022/071328, Sep. 9, 2022. cited by applicant

International Search Report and Written Opinion for PCT/US2022/071329, Aug. 25, 2022. cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/305,297, filed Jul. 2, 2021, titled “SYSTEMS FOR GENERATING GEOTHERMAL POWER IN AN ORGANIC RANKINE CYCLE OPERATION DURING HYDROCARBON PRODUCTION BASED ON WORKING FLUID TEMPERATURE”, which claims priority to and the benefit of U.S. Provisional Application No. 63/200,908, filed Apr. 2, 2021, titled “SYSTEMS AND METHODS FOR GENERATING GEOTHERMAL POWER DURING HYDROCARBON PRODUCTION,” the disclosures of which are incorporated herein by reference in their entirety.

FIELD OF DISCLOSURE

(1) Embodiments of this disclosure relate to generating geothermal power during hydrocarbon production, and more particularly, to systems and methods for generating and controllers for controlling generation of geothermal power in an organic Rankine cycle (ORC) operation in the vicinity of a wellhead during hydrocarbon production to thereby supply electrical power to one or

more of in-field operational equipment, a grid power structure, and an energy storage device.

BACKGROUND

(2) Typically, geothermal generators include a working fluid loop that flows to varying depths underground, such that underground heat causes the working fluid in the loop to change phases from a liquid to a vapor. The vaporous working fluid may then flow to a gas expander, causing the gas expander to rotate. The rotation of the gas expander may cause a generator to generate electrical power. The vaporous working fluid may then flow to a condenser or heat sink. The condenser or heat sink may cool the working fluid, causing the working fluid to change phase from the vapor to the liquid. The working fluid may circulate through the loop in such a continuous manner, thus the geothermal generator may generate electrical power.

(3) Heat exchangers within geothermal generators are not built to withstand high pressures. Typically, the working fluid does not flow through the loop under high pressure. Further, rather than including heat exchangers, typically, geothermal generators utilize geothermal heat from varying depths underground, the heat being transferred to the working fluid. For example, a geothermal generator may simply include a conduit or pipe buried deep underground. As working fluid flows through the conduit or pipe, the heat at such depths causes the working fluid to change to a vapor and the vapor may flow up to the gas expander. As the vapor is cooled, the vapor flows back underground, the cycle repeating. While old or non-producing wells have been utilized for geothermal power generation, no such solution is known to exist for generating geothermal power at a well during hydrocarbon production.

(4) Accordingly, Applicants have recognized a need for systems and methods to generate geothermal power in an organic Rankine cycle (ORC) operation in the vicinity of a wellhead during hydrocarbon production to thereby supply electrical power to one or more of in-field operational equipment, a grid power structure, and an energy storage device. The present disclosure is directed to embodiments of such systems and methods

SUMMARY

(5) The present disclosure is generally directed to systems and methods for generating and controlling generation of geothermal power in an organic Rankine cycle (ORC) operation in the vicinity of a wellhead during hydrocarbon production to thereby supply electrical power to one or more of in-field operational equipment, a grid power structure, and an energy storage device. The wellhead may produce a wellhead fluid. The wellhead fluid, as it exits the wellhead, may be under high-pressure and at a high temperature. The wellhead fluid's temperature may be determined, as well as the pressure, and based on such determinations, one or more heat exchanger valves may be actuated to partially open or completely open, thereby diverting a portion or all of the flow of wellhead fluid to the heat exchanger. The heat exchanger may be a high-pressure heat exchanger configured to withstand the high-pressure of the wellhead fluid from a wellhead. The heat exchanger may indirectly transfer heat from the flow of the wellhead fluid to the flow of a working fluid. As heat is transferred from the flow of the wellhead fluid to the flow of a working fluid, such a heat transfer may cause the working fluid to change phases from a liquid to a vapor. The vaporous working fluid may then flow through an ORC unit to cause a generator to generate electrical power via rotation of a gas expander of the ORC unit. Such an operation may be defined as or may be an ORC operation or process. The ORC unit may be an off-the-shelf unit, while the high-pressure heat exchanger may be a stand-alone component or device. In another embodiment, the ORC unit may be a high-pressure ORC unit and may include the high-pressure heat exchanger apparatus.

(6) Accordingly, an embodiment of the disclosure is directed to a system for generating geothermal power in an organic Rankin cycle (ORC) operation in the vicinity of a wellhead during hydrocarbon production to thereby supply electrical power to one or more of in-field equipment, a grid power structure, and energy storage devices. The system may include a first temperature sensor to provide a first temperature, the first temperature defined by a temperature of a flow of

organic working fluid. The system may include a heat exchanger valve to divert flow of wellhead fluid from one or more wellheads based on the first temperature. The system may include a high-pressure heat exchanger including a first fluid path to accept and output the flow of wellhead fluid from the heat exchanger valve and a second fluid path to accept and output the flow of the organic working fluid, the high-pressure heat exchanger to indirectly transfer heat from the flow of wellhead fluid to the flow of the organic working fluid to cause the organic working fluid to change phases from a liquid to a vapor. The system may include an ORC unit. The ORC unit may include a generator, a gas expander, and a partial loop for the flow of the organic working fluid, the partial loop defined by a fluid path through the condenser, generator, and pump, the partial loop forming a complete loop when connected to the fluid path of the high-pressure heat exchanger, the flow of the organic working fluid, as a vapor, so as to cause the generator to generate electrical power via rotation of a gas expander as defined by an ORC operation, the condenser to cool the flow of the organic working fluid, the cooling to cause the organic working fluid to change phases from the vapor to the liquid, the pump to transport the liquid state organic working fluid from the condenser for heating.

(7) In another embodiment, the system may include a pump and/or a condenser. The system may include a first wellhead fluid valve to adjust flow of wellhead fluid from the one or more wellheads based on the diversion of the flow of wellhead fluid to the heat exchanger valve. The system may include another one or more ORC units connected to the high-pressure heat exchanger. The ORC unit may include a working fluid reservoir to store organic working fluid flowing from the condenser.

(8) In another embodiment, the first fluid path of the high-pressure heat exchanger may be configured to withstand corrosion caused by the wellhead fluid. The high-pressure heat exchanger may include a vibration induction device to reduce scaling caused by the flow of the wellhead fluid.

(9) Another embodiment is directed to a system for generating geothermal power in the vicinity of a wellhead during hydrocarbon production to thereby supply electrical power to one or more of in-field equipment, a grid power structure, and energy storage devices. The system may include a first pipe connected to and in fluid communication with the wellhead. The first pipe may be configured to transport wellhead fluid under high-pressure. The system may include a first wellhead fluid valve having a first end and second end, the first end of the first wellhead fluid valve connected to and in fluid communication with the first pipe, the first wellhead fluid valve to control flow of wellhead fluid based on an organic working fluid temperature. The system may include a heat exchanger valve connected to and in fluid communication with the first pipe, the heat exchanger valve to control flow of wellhead fluid on an organic working fluid temperature. The system may include a high-pressure heat exchanger to accept the flow of wellhead fluid when the heat exchanger valve is open. The high-pressure heat exchanger may include a first opening and a second opening connected via a first fluidic path and a third opening and a fourth opening connected via a second fluidic path, the first fluidic path and the second fluidic path to facilitate heat transfer from the flow of wellhead fluid to an organic working fluid, the transfer of heat from the wellhead fluid to the organic working fluid to cause the organic working fluid to change phases from a liquid to a vapor, the flow of wellhead fluid flowing into the first opening of the high-pressure heat exchanger from the heat exchanger valve through the first fluidic path and to the second opening of the high-pressure heat exchanger, and a flow of the organic working fluid flowing into the third opening through the second fluidic path and out of the fourth opening. The system may include a first temperature sensor connected to the fourth opening, the first temperature sensor to provide the organic working fluid temperature, the organic working fluid temperature defined by a temperature of organic working fluid flowing through the second fluidic path. The system may include a second pipe connected to and in fluid communication with the second end of the first wellhead fluid valve and connected to and in fluid communication with the second opening

of the high-pressure heat exchanger. The system may include a generator connected to and in fluid communication with the fourth opening of the high-pressure heat exchanger, the organic working fluid flowing from the fourth opening to the generator and causing the generator to generate electrical power via rotation of a vapor expander as defined by an ORC operation. The system may include a condenser to facilitate heat transfer from the organic working fluid to a coolant. The system may include a pump connected to the condenser to pump the organic working fluid from the condenser to the third opening of the high-pressure heat exchanger.

(10) In an embodiment, the organic working fluid may include one of pentafluoropropane, carbon dioxide, ammonia and water mixtures, tetrafluoroethane, isobutene, propane, pentane, perfluorocarbons, and other hydrocarbons. The generator may include a rotation mechanism, a stator, and rotor, the rotor connected to the rotation mechanism, the rotor to rotate as the rotation mechanism spins via the flow of organic working fluid. The rotation mechanism may include one of a turbine expander, a positive displacement expander, or a twin-screw expander. The rotation mechanism may connect to the rotor via one of a transmission and gearbox.

(11) In another embodiment, when the heat exchanger valve is opened, a portion of the flow of wellhead fluid is diverted from the first wellhead fluid valve. The organic working fluid may change phase from liquid to vapor when the wellhead fluid is at a temperature from greater than or equal to about 50 degrees Celsius.

(12) In another embodiment, after the organic working fluid flows through the expander, the organic working fluid flows through a regenerator and, after the working fluid flows through the condenser, the organic working fluid flows through the regenerator, the regenerator pre-heating the organic working fluid from the condenser via heat of the organic working fluid from the generator.

(13) In another embodiment, the high-pressure heat exchanger may be configured to withstand pressures up to about 15,000 pounds per square inch (PSI). The high-pressure heat exchanger may include pressure relief valves, in an event that the wellhead fluid flowing to the high-pressure heat exchanger exceeds a pressure rating of the high-pressure heat exchanger.

(14) In another embodiment, the wellhead fluid includes one of a liquid and a gas. The wellhead fluid may include hydrocarbons. The wellhead fluid may further include a mixture of the hydrocarbons and one or more of water and other chemical residuals. The working fluid may include a mixture of two or more fluids, each of the two or more fluids including different vaporous phase change points and different condensation points.

(15) In another embodiment, the generator, condenser, and pump define an ORC unit. The system may include another one or more high-pressure heat exchangers. The ORC unit may connect to each of the high-pressure heat exchanger and the another one or more high-pressure heat exchangers. Each of one or more ORC unit valves may connect and be positioned between the ORC unit and each of the high-pressure heat exchanger and the another one or more high-pressure heat exchangers. Each of the one or more ORC unit valves may be adjustingly opened to a position based on an efficient working fluid flow to the ORC unit. The efficient working fluid flow may be based on one or more of a temperature of working fluid flowing from each of the high-pressure heat exchanger and the another one or more high-pressure heat exchangers, an electrical output from the ORC unit, and a current open position of each of the one or more ORC unit valves.

(16) Still other aspects and advantages of these embodiments and other embodiments, are discussed in detail herein. Moreover, it is to be understood that both the foregoing information and the following detailed description provide merely illustrative examples of various aspects and embodiments, and are intended to provide an overview or framework for understanding the nature and character of the claimed aspects and embodiments. Accordingly, these and other objects, along with advantages and features of the present invention herein disclosed, will become apparent through reference to the following description and the accompanying drawings. Furthermore, it is to be understood that the features of the various embodiments described herein are not mutually exclusive and may exist in various combinations and permutations.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) These and other features, aspects, and advantages of the disclosure will become better understood with regard to the following descriptions, claims, and accompanying drawings. It is to be noted, however, that the drawings illustrate only several embodiments of the disclosure and, therefore, are not to be considered limiting of the scope of the disclosure.
- (2) FIG. 1A and FIG. 1B are schematic top-down perspectives of novel implementations of a geothermal power generation enabled well, according to one or more embodiment of the disclosure.
- (3) FIG. 2A, FIG. 2B, FIG. 2C, FIG. 2D, FIG. 2E, FIG. 2F, FIG. 2G, and FIG. 2H are block diagrams illustrating novel implementations of a geothermal power generation enabled well to provide electrical power to one or more of in-field equipment, equipment at other wells, energy storage devices, and the grid power structure, according to one or more embodiment of the disclosure.
- (4) FIG. 3A and FIG. 3B are block diagrams illustrating other novel implementations of a geothermal power generation enabled well to provide electrical power to one or more of in-field equipment, equipment at other wells, energy storage devices, and the grid power structure, according to one or more embodiment of the disclosure.
- (5) FIG. 4A and FIG. 4B are simplified diagrams illustrating a control system for managing geothermal power production at a well, according to one or more embodiment of the disclosure.
- (6) FIG. 5 is a flow diagram of geothermal power generation in which, when a wellhead fluid is at or above a vaporous phase change temperature, heat exchanger valves may be opened to allow wellhead fluid to flow therethrough, thereby facilitating heating of a working fluid for use in an organic Rankine cycle (ORC) unit, according to one or more embodiment of the disclosure.
- (7) FIG. 6 is another flow diagram of geothermal power generation in which, when a wellhead fluid is at or above a vaporous phase change temperature, heat exchanger valves may be opened to allow wellhead fluid to flow therethrough, thereby facilitating heating of a working fluid for use in a ORC unit, according to one or more embodiment of the disclosure.
- (8) FIG. 7A is a flow diagram of geothermal power generation in which, when an ORC fluid is at or above a vaporous phase change temperature, heat exchanger valves may remain open to allow wellhead fluid to flow therethrough, thereby facilitating heating of a working fluid for use in an ORC unit, according to one or more embodiment of the disclosure.
- (9) FIG. 7B is another flow diagram of geothermal power generation in which, when an ORC fluid is at or above a vaporous phase change temperature, heat exchanger valves may remain open to allow wellhead fluid to flow therethrough, thereby facilitating heating of a working fluid for use in an ORC unit, according to one or more embodiment of the disclosure.
- (10) FIG. 8 is a flow diagram of geothermal power generation in which a working fluid flow is determined based on a preselected electrical power output threshold, according to one or more embodiment of the disclosure.
- (11) FIG. 9A, FIG. 9B, FIG. 9C, FIG. 9D, FIG. 9E, FIG. 9F, and FIG. 9G are block diagrams illustrating novel implementations of one or more geothermal power generation enabled wells to provide electrical power to one or more of in-field equipment, equipment at one of the other wells, energy storage devices, and the grid power structure, according to one or more embodiment of the disclosure.

DETAILED DESCRIPTION

- (12) So that the manner in which the features and advantages of the embodiments of the systems and methods disclosed herein, as well as others that will become apparent, may be understood in more detail, a more particular description of embodiments of systems and methods briefly

summarized above may be had by reference to the following detailed description of embodiments thereof, in which one or more are further illustrated in the appended drawings, which form a part of this specification. It is to be noted, however, that the drawings illustrate only various embodiments of the systems and methods disclosed herein and are therefore not to be considered limiting of the scope of the systems and methods disclosed herein as it may include other effective embodiments as well.

(13) The present disclosure is directed to systems and methods for generating geothermal power in an organic Rankine cycle (ORC) operation in the vicinity of a wellhead during hydrocarbon production to thereby supply electrical power to one or more of in-field equipment or operational equipment, a grid power structure, other equipment, and an energy storage device. Wellhead fluids flowing from a wellhead at a well are typically under high-pressure. In-field equipment at the well is not rated for such high pressures. Prior to further processing or transport, the pressure of the flow of the wellhead fluid may be reduced, e.g., from 15,000 PSI to 200 PSI, from 10,000 PSI to 200 PSI, from 2,000 PSI to 200 PSI, or any other range from 20,000 PSI to 100 PSI, based on the pressure rating of the in-field equipment at the well. As the wellhead fluid flows from the wellhead, the temperature of the flow of the wellhead fluid may be at a high temperature, at least partially due to the high pressure of the flow of the wellhead fluid. As the pressure is reduced, the wellhead fluid temperature may also be reduced, as result of the pressure drop. Typically, the heat of the flow of wellhead fluid from the wellhead is not utilized and may be considered heat waste.

(14) Geothermal power generators typically use a looping pipe or pipeline buried at depths with sufficient temperature to allow a working fluid to change phase from liquid to vapor. As the working fluid changes phase from a liquid to a vaporous state, the vaporous state working fluid may flow up the pipe or pipeline to a gas expander. The vaporous state working fluid may flow through and cause the gas expander to rotate. The rotation of the gas expander may cause a generator to generate electrical power, as will be described below. The vaporous state working fluid may flow through the gas expander to a heat sink, condenser, or other cooling apparatus. The heat sink, condenser, or other cooling apparatus may cool the working fluid thereby causing the working fluid to change phases from a vapor to a liquid. Heat exchangers of typical geothermal generators are not rated for high-pressure operations and usually geothermal generators obtain heat from varying underground depths.

(15) In the present disclosure, a high-pressure heat exchanger may be placed or disposed at the well and/or in the vicinity of one or more wellheads. The high-pressure heat exchanger may be connected to the wellhead and may accept a high temperature or heated flow of wellhead fluid. A working fluid may flow through the heat exchanger. As the wellhead fluid and working fluid flows through the high-pressure heat exchanger, the high-pressure heat exchanger may facilitate transfer of heat from the wellhead fluid to the working fluid. A heat exchanger may include two fluidic paths, one for a heated fluid and another for a cool fluid. The fluidic paths may be in close proximity, allowing heat to transfer from the heated fluid to the cool fluid. The fluidic paths may be loops, coils, densely packed piping, tubes, chambers, some other type of path to allow for fluid to flow therethrough, and/or a combination thereof, as will be understood by those skilled in the art. As fluids flow through the heat exchanger, the cool liquid's temperature may increase, while the heated liquid's temperature may decrease.

(16) Additionally, a geothermal generator unit or ORC unit may be disposed, positioned, or placed at the wellhead. The geothermal generator unit or ORC unit may directly connect to the high-pressure heat exchanger, include the high-pressure heat exchanger, or may connect to the high-pressure heat exchanger via an intermediary heat exchanger. As the hot wellhead fluid heats the working fluid, either via direct connection or through an intermediary heat exchanger, the working fluid may change phases from a liquid to a vapor. In such examples, the working fluid utilized may be chosen based on a low boiling point and/or high condensing point. The vaporous state working fluid may flow through the geothermal generator unit or ORC unit to a generator, e.g., a gas

expander and generator. The vaporous state working fluid may then flow to a condenser or heat sink, thereby changing state from the vapor to the liquid. Finally, the liquid may be pumped back to the high-pressure heat exchanger. Such a cycle, process, or operation may be considered a Rankine cycle or ORC.

(17) Such systems may include various components, devices, or apparatuses, such as temperature sensors, pressure sensors or transducers, flow meters, control valves, smart valves, valves actuated via control signal, controllers, a master or supervisory controller, other computing devices, computing systems, user interfaces, in-field equipment, and/or other equipment. The controller may monitor and adjust various aspects of the system to ensure that hydrocarbon production continues at a specified rate, that downtime is limited or negligible, and that electrical power is generated efficiently, optimally, economically, and/or to meet or exceed a preselected electrical power output threshold.

(18) FIGS. 1A and 1B are schematic top-down perspectives of novel implementations of a geothermal power generation enabled well, according to one or more embodiment of the disclosure. As illustrated in FIG. 1A, a well **100** may include various components or equipment, also referred to as in-field equipment. Such in-field equipment may include fracturing equipment, field compressors, pump stations, artificial lift equipment, drilling rigs, data vans, and/or any other equipment utilized or used at a well **100**. For example, the well **100** may include one or more pumpjacks **108**, one or more wellhead compressors **110**, various other pumps, various valves, and/or other equipment that may use electrical power or other type of power to operate. To generate power from otherwise wasted heat, the well **100** may additionally include a high-pressure heat exchanger **104**, one or more geothermal generators, one or more ORC units **106**, a high-pressure geothermal generator, a high-pressure ORC unit, and/or some combination thereof. As wellhead fluid flows from one of the one or more wellheads **102**, a portion of the flow of wellhead fluid or all of the flow of wellhead fluid may flow through the high-pressure heat exchanger **104**, high-pressure geothermal generator, a high-pressure ORC unit, or some combination thereof. As the hot and high-pressure wellhead fluid flows through, for example, the high-pressure heat exchanger **104**, the high-pressure heat exchanger **104** may facilitate a transfer of heat from the wellhead fluid to a working fluid flowing through the high-pressure heat exchanger **104**. In other words, the wellhead fluid may heat the working fluid. Such a heat transfer may cause the working fluid to change phases from a liquid to a vapor. The vaporous state working fluid may flow from the high-pressure heat exchanger to the one or more ORC units **106**. The one or more ORC units **106** may then generate power using the vaporous state working fluid. The electrical power may be transferred to the in-field equipment at the well **100**, to an energy storage device (e.g., if excess power is available), to equipment at other wells, to the grid or grid power structure (e.g., via a transformer **116** through power lines **118**), or some combination thereof.

(19) As illustrated in FIG. 1B, one or more wells **100A**, **100B**, **100C** may be nearby or in close proximity to each of the other one or more wells **100A**, **100B**, **100C**. Further, each of the one or more wells **100A**, **100B**, **100C** may utilize different amounts of electrical power, in addition to generating different amounts of electrical power. As such, one well (e.g., well **100A**, well **100B**, and/or well **100C**) of the one or more wells **100A**, **100B**, **100C** may generate a surplus of electrical power or utilize electrical power from other sources. In an example, a controller may determine if a well (e.g., well **100A**, well **100B**, and/or well **100C**) of the one or more wells **100A**, **100B**, **100C** generates a surplus. If a surplus is generated, the controller may determine which, if any, of the other one or more wells **100A**, **100B**, **100C** may have a deficit of electrical power. The controller may then transmit signals to equipment at the one or more wells **100A**, **100B**, **100C** to enable electrical power transfer between the one or more wells **100A**, **100B**, **100C** with excess and deficits, e.g., a well with a deficit may receive electrical power from a well with a surplus (see **120**). In another example, the one or more wells **100A**, **100B**, **100C** may include energy storage devices e.g., batteries, battery banks, or other solutions to store energy for short or long term time

periods. The energy storage devices may be placed, disposed, or installed at one or more of the one or more wells **100A**, **100B**, **100C** or at points in between the one or more wells **100A**, **100B**, **100C**. As surplus electrical power is generated, that surplus electrical power may be transmitted and stored in the energy storage devices. The energy storage devices may be accessible by the in-field equipment of each of the one or more wells **100**.

(20) As illustrated in FIGS. **1A** and **1B**, the one or more wells **100A**, **100B**, **100C** may include a high-pressure heat exchanger **104** and ORC units **106**. In an example, the ORC units **106** may be modular and/or mobile. The ORC units **106** may be mounted to a vehicle, such as a truck or other vehicle type, or skid and transported to the well **100**. Further, the high-pressure heat exchanger **104** may be modular and/or mobile. The high-pressure heat exchanger **104** may be mounted to a vehicle and/or skid. Upon arrival at one of the one or more wells **100A**, **100B**, **100C**, the high-pressure heat exchanger **104** may be removed from the vehicle or the vehicle may be left on-site or at least while the one or more wells **100A**, **100B**, **100C** are producing hydrocarbons. In an example, during hydrocarbon production, operation of the high-pressure heat exchanger **104** and ORC unit **106** may occur. After hydrocarbon production has ceased, none, some, or all of the equipment may be removed from the well **100**. For example, the well **100** may be re-used for generating geothermal energy via a different method. In such examples the ORC units **106** and high-pressure heat exchanger **104** may remain on-site.

(21) FIG. **2A**, FIG. **2B**, FIG. **2C**, FIG. **2D**, FIG. **2E**, FIG. **2F**, FIG. **2G**, and FIG. **2H** are block diagrams illustrating novel implementations of a geothermal power generation enabled well to provide electrical power to one or more of in-field equipment, equipment at other wells, energy storage devices, and the grid power structure, according to one or more embodiment of the disclosure. As illustrated in the FIGS. **2A** through **2H**, different embodiments may be utilized for geothermal power generation at the surface of a well during hydrocarbon production. As illustrated in FIG. **2A**, a well **200** may include a wellhead **202**. The wellhead **202** may produce a stream or flow of wellhead fluid. The wellhead fluid may flow from a first pipe **215**. The first pipe **215** may include, at various positions along the length of the pipe **215**, sensors, meters, transducers, and/or other devices to determine the characteristics of the wellhead fluid flowing through pipe **215**. Further, sensors, meters, transducers and/or other devices may be included at various positions within the high-pressure heat exchanger **250** and/or ORC unit **203** to determine the characteristics of the working fluid or ORC fluid flowing through the high-pressure heat exchanger **250** and/or ORC unit **203**. Based on the determined characteristics of the wellhead fluid, working fluid, and/or ORC fluid and/or characteristics of other aspects of the well **200**, e.g., temperature, pressure, flow, power demand and/or power storage or transmission options and/or other factors, a first heat exchanger valve **208** connected to the first pipe **215** may be opened, e.g., partially opened or fully opened.

(22) Further, when the first heat exchanger valve **208** is opened, a second heat exchanger valve **222** connected to a second pipe **217** may be opened, e.g., partially opened or fully opened. The second heat exchanger valve **222** may allow the flow of wellhead fluid to exit the high-pressure heat exchanger **250**. Based on the opening of the first heat exchanger valve **208** and second heat exchanger valve **222**, a first wellhead fluid valve **210** may be adjusted. Such adjustment may occur to ensure that the production of the wellhead **202** or the production of wellhead fluid from the wellhead **202** may not be impeded or slowed based on a diversion of the flow of the wellhead fluid to the high-pressure heat exchanger **250**. Downstream of the first wellhead fluid valve **210**, but prior to where the diverted portion of the flow of wellhead fluid is reintroduced to the primary or bypass wellhead fluid flow, a pressure sensor **212** may be disposed, e.g., the pressure sensor **212** may be disposed along pipe **217**. In another example, rather than a pressure sensor **212**, a flow meter may be disposed along pipe **217**, e.g., a wellhead fluid flow meter and/or a downstream flow meter. The wellhead fluid flow meter may measure the flow rate of a fluid exiting the wellhead **202**. A downstream flow meter may measure the flow rate of the wellhead fluid at a point

downstream of the first wellhead fluid valve **210**. Such a pressure sensor **212** or flow meter may be utilized to determine whether the flow of wellhead fluid is at a pressure or flow that does not impede hydrocarbon production. In an example, where the flow is impeded to a degree that the production of hydrocarbons may be inhibited, the first wellhead fluid valve **210** may be opened further, if the first wellhead fluid valve **210** is not already fully opened. If the first wellhead fluid valve **210** is fully opened, the first heat exchanger valve's **208** and second heat exchanger valve's **222** percent open may be adjusted. Other factors may be taken into account for such determinations, such as the pressure or flow from the high-pressure heat exchanger **250**, the pressure or flow at the point of a choke valve located further downstream along the second pipe **217**, a pressure rating for downstream equipment, and/or the temperature of the wellhead fluid within the high-pressure heat exchanger **250**. Once the flow of wellhead fluid passes through the first wellhead fluid valve **210** and/or the second heat exchanger valve **222**, the wellhead fluid may flow to in-field equipment **270** for further processing or transport.

(23) Each of the valves described herein, e.g., first heat exchanger valve **208**, second heat exchanger valve **222**, first wellhead fluid valve **210**, and other valves illustrated in the FIG. 2B through FIGS. 9G, may be control valves, electrically actuated valves, pneumatic valves, or other valves suitable to receive signals and open and close under, potentially, high pressure. The valves may receive signals from a controller or other source and the signals may cause the valves to move to a partially or fully opened or closed position. The signal may indicate the position that the valve may be adjusted to, e.g., a position halfway open, a position a third of the way open, a position a particular degree open, or completely open, such positions to be understood as non-limiting examples. In such examples, as the valve receives the signal indicative of a position to adjust to, the valve may begin turning to the indicated position. Such an operation may take time, depending on the valve used. To ensure proper operation and prevent damage (e.g., damage to the high-pressure heat exchanger **250**, such as when pressure of the wellhead fluid exceeds a pressure rating of the high-pressure heat exchanger **250**), the valve may be configured to close in a specified period of time while high-pressure fluid flows therethrough. Such a configuration may be based on a torque value of the valve, e.g., a valve with a higher torque value may close faster than that of a control valve with a lower torque value. Such a specified period of time may be 5 seconds to 10 seconds, 5 seconds to 15 seconds, 5 seconds to 20 seconds, 10 seconds to 15 seconds, 10 seconds to 20 seconds, or 15 seconds to 20 seconds. For example, if a wellhead fluid flow exceeds a pressure of 15,000 PSI, the first heat exchanger valve **208** may close within 5 seconds of the pressure sensor **214** indicating the pressure exceeding 15,000 PSI. In other embodiments, the valves may either fully open or fully close, rather than open to positions in between. In yet other examples, the valves may be manually or physically opened by operators or technicians on-site.

(24) As illustrated in FIG. 2A, pressure sensors **204**, **212**, **214**, **222**, **207** and/or temperature sensors **206**, **216**, **218**, **226**, **218** may be disposed at various points on or along different pipes, equipment, apparatuses, and/or devices at the well. Each sensor may provide information to adjust and/or control various aspects of the wellhead fluid flow. For example, pressure sensors **204**, **212** may provide pressure measurements to determine whether a flow of wellhead fluid is not impeded in relation to hydrocarbon production targets. Pressure sensors **214**, **220** may provide measurements to ensure that the flow of wellhead fluid through the high-pressure heat exchanger **250** is sufficient to facilitate heat transfer from the wellhead fluid to a working fluid in a ORC loop **221** and/or that the pressure within the high-pressure heat exchanger **250** does not exceed a pressure rating of the high-pressure heat exchanger **250**. In another example, temperature sensor **206** may provide data to control flow of the wellhead fluid. For example, if the flow of wellhead fluid is at a temperature sufficient to cause the working fluid to exhibit a vaporous phase change, then the first heat exchanger valve **208** and second heat exchanger valve **222** may be opened. Further, temperature sensors **216**, **218**, **226**, **248** may provide measurements to ensure that the temperatures within the heat exchanger, of the flow of wellhead fluid, and of the flow of working fluid, are above the

thresholds or within a range of thresholds, e.g., above a vaporous phase change threshold. For example, rather than or in addition to using temperature of the wellhead fluid as measured by temperature sensor **206**, the temperature of the working fluid as measured by temperature sensor **226** may be utilized to determine whether to maintain an open position of an already open first heat exchanger valve **208**. In such examples, the first heat exchanger valve **208** may initially be fully or partially open, e.g., prior to flow of the wellhead fluid.

(25) In another embodiment, in addition to the first heat exchanger valve **208** and second heat exchanger valve **222**, the well **200** may include a first ORC unit valve **295** and a second ORC unit valve **296**. Such ORC unit valves **295**, **296** may be utilized to control flow of working fluid flowing into the ORC unit **203**. Further the ORC unit valves **295**, **296** may be utilized when more than one high-pressure heat exchanger **250** corresponding to one or more wellheads are connected to the ORC unit **203**. The ORC unit valves **295**, **296** may be utilized to optimize or to enable the ORC unit **203** to meet a preselected electrical power output threshold via the flow of working fluid from one or more high-pressure heat exchangers into the ORC unit **203**, as will be understood by a person skilled in the art. The flow of working fluid may be adjusted to ensure that the ORC unit **203** produces an amount of electrical power greater than or equal to a preselected electrical power output threshold, based on various factors or operating conditions (e.g., temperature of wellhead fluid flow, temperature of working fluid flow, electrical output **236** of the ORC unit **203**, electrical rating of the ORC unit **203**, flow and/or pressure of the wellhead fluid, and/or flow and/or pressure of the working fluid). In some examples, the ORC unit valves **295**, **296** may initially be fully open or at least partially open and as various factors or operating conditions are determined, then the ORC unit valves **295**, **296** for one or more heat exchangers may be adjusted to enable the ORC unit **203** to meet a preselected electrical power output threshold, as will be understood by a person skilled in the art. The preselected electrical power output threshold may be set by a user or may be a predefined value generated by a controller (e.g., controller **272**) based on various factors. The various factors may include an ORC unit electrical power rating or output rating or maximum potential temperature of the wellhead fluid and/or working fluid.

(26) As shown, several pairs of sensors may be located adjacent to one another. In other examples, those positions, for example, the pressure sensor's **204** and the temperature sensor's **206** location, may be reversed. In yet another example, each one of the sensors may provide measurements for multiple aspects of the wellhead fluid, e.g., one sensor to provide a combination of flow, pressure, temperature, composition (e.g., amount of components in the wellhead fluid, such as water, hydrocarbons, other chemicals, proppant, etc.), density, or other aspect of the wellhead fluid or working fluid. Each sensor described above may be integrated in or within the pipes or conduits of each device or component, clamped on or over pipes or conduits, and or disposed in other ways, as will be understood by those skilled in the art. Further, the determinations, adjustments, and/or other operations described above may occur or may be performed by or in a controller.

(27) As noted, a high-pressure heat exchanger **250** may be disposed, placed, or installed at a well. The high-pressure heat exchanger **250** may be disposed nearby or at a distance from the wellhead **202**. The high-pressure heat exchanger **250** may be a modular and/or mobile apparatus. In such examples, the high-pressure heat exchanger **250** may be brought or moved to a well or site (e.g., via a vehicle, such as a truck), placed at the well or site during hydrocarbon production, and then moved to another well or site at the end of hydrocarbon production. The high-pressure heat exchanger **250** may be disposed on a skid, a trailer, a flatbed truck, inside a geothermal generator unit, or inside an ORC unit **203**. Once brought to a well or site, the high-pressure heat exchanger **250** may be secured to the surface at the well. The high-pressure heat exchanger **250** may be configured to withstand pressures in excess of about 5,000 PSI, about 10,000 PSI, about 15,000 PSI, and/or greater. In an example, the high-pressure heat exchanger **250** may be a high-pressure shell and tube heat exchanger, a spiral plate or coil heat exchanger, a heliflow heat exchanger, or other heat exchanger configured to withstand high pressures. In another example, portions of the

high-pressure heat exchanger **250** may be configured to withstand high-pressures. For example, if a shell and tube heat exchanger is utilized, the shell and/or tubes may be configured to withstand high-pressures.

(28) In another embodiment, at least one fluidic path of the high-pressure heat exchanger **250** may be coated or otherwise configured to reduce or prevent corrosion. In such examples, a wellhead fluid may be corrosive. To prevent damage to the high-pressure heat exchanger **250** over a period of time, the fluid path for the wellhead fluid may be configured to withstand such corrosion by including a permanent, semi-permanent, or temporary anti-corrosive coating, an injection point for anti-corrosive chemical additive injections, and/or some combination thereof. Further, at least one fluid path of the high-pressure heat exchanger **250** may be comprised of an anti-corrosive material, e.g., anti-corrosive metals or polymers. As noted, the wellhead fluid may flow into the high-pressure heat exchanger **250** at a high pressure. As the high-pressure heat exchanger **250** may operate at high pressure, the high-pressure heat exchange may include pressure relief valves to prevent failures if pressure within the high-pressure heat exchanger **250** were to exceed the pressure rating of the high-pressure heat exchanger **250**. Over time, wellhead fluid flowing through the high-pressure heat exchanger **250** may cause a buildup of deposits or scaling. To prevent scaling and/or other related issues, the high-pressure heat exchanger **250** may be injected with scaling inhibitors or other chemicals or may include vibration or radio frequency induction devices.

(29) Once the high-pressure heat exchanger **250** facilitates heat transfer from the wellhead fluid to the working fluid, the working fluid may partially, substantially, or completely change phases from a liquid to a vapor, vaporous state, gas, or gaseous state. The vapor or gas may flow to the ORC unit **203** causing an expander to rotate. The rotation may cause a generator to generate electricity, as will be further described and as will be understood by those skilled in the art. The generated electricity may be provided as an electrical output **236**. The electricity generated may be provided to in-field equipment, energy storage devices, equipment at other wells, or to a grid power structure. The working fluid in the high-pressure heat exchanger may be a working fluid to carry heat. Further, the working fluid of the high-pressure heat exchanger **250** may or may not exhibit a vaporous phase change. The working fluid may carry heat to another heat exchanger **205** of the ORC unit **203**. As such, heat may be transferred from the wellhead fluid to the working fluid of the high-pressure heat exchanger **250** and heat may be transferred from the working fluid of the high-pressure heat exchanger **250** to the working fluid of the ORC unit **203**.

(30) In an example, the working fluid may be a fluid with a low boiling point and/or high condensation point. In other words, a working fluid may boil at lower than typical temperatures, while condensing at higher than typical temperatures. The working fluid may be an organic working fluid. The working fluid may be one or more of pentafluoropropane, carbon dioxide, ammonia and water mixtures, tetrafluoroethane, isobutene, propane, pentane, perfluorocarbons, other hydrocarbons, a zeotropic mixture of pentafluoropentane and cyclopentane, other zeotropic mixtures, and/or other fluids or fluid mixtures. The working fluid's boiling point and condensation point may be different depending on the pressure within the ORC loop **221**, e.g., the higher the pressure, the lower the boiling point.

(31) FIG. 2B illustrates an embodiment of the internal components of an ORC unit **203**. FIG. 2B further illustrates several of the same components, equipment, or devices as FIG. 2A illustrates. As such, the numbers used to label FIG. 2B may be the same as those used in 2A, as those numbers correspond to the same component. The ORC unit **203** as noted may be a modular or mobile unit. As power demand increases, additional ORC units **203** may be added, installed, disposed, or placed at the well. The ORC units **203** may stack, connect, or integrate with each other ORC unit. In an example, the ORC unit **203** may be a modular single-pass ORC unit.

(32) An ORC unit **203** may include a heat exchanger **205** or heater. Connections to the heat exchanger **205** or heater may pass through the exterior of the ORC unit **203**. Thus, as an ORC unit **203** is brought or shipped to a well or other location, a user, technician, service person, or other

person may connect pipes or hoses from a working fluid heat source (e.g., the high-pressure heat exchanger **250**) to the connections on the ORC unit **203**, allowing a heat source to facilitate phase change of a second working fluid in the ORC unit **203**. In such examples, the working fluid flowing through ORC loop **221** may include water or other organic fluid exhibiting a higher vaporous phase change threshold than the working fluid of the ORC unit **203**, to ensure proper heat transfer in heat exchanger **205**. Further, the heat exchanger **205** may not be a high-pressure heat exchanger. In such examples, the high-pressure heat exchanger **250** allows for utilization of waste heat from high-pressure wellhead fluids. In another embodiment and as will be described, a high-pressure heat exchanger **250** may be included in the ORC unit **203**.

(33) In yet another embodiment, the high-pressure heat exchanger **250** may be considered an intermediary heat exchanger or another intermediary heat exchanger (e.g., intermediary heat exchanger **219**) may be disposed between the high-pressure heat exchanger **250** and the ORC unit **203** (as illustrated in FIG. 2C and described below). The working fluid flowing through the high-pressure heat exchanger **250** may be sufficient to heat another working fluid of the ORC unit **203**. The working fluid of such an intermediary heat exchanger may not physically flow through any of the equipment in the ORC unit **203**, except for heat exchanger **205**, thereby transferring heat from the working fluid in ORC loop **221** to an ORC working fluid in a loop defined by the fluid path through the heat exchanger **205**, condenser **211**, expander **232**, working fluid reservoir **298**, and/or pump **244**.

(34) The ORC unit **203** may further include pressure sensors **228**, **238**, **246** and temperature sensors **230**, **240** to determine whether sufficient, efficient, and/or optimal heat transfer is occurring in the heat exchanger **205**. A sensor or meter may further monitor electrical power produced via the expander **232** and generator **234**. Further, the ORC unit **203** may include a condenser or heat sink **211** to transfer heat from the second working fluid or working fluid of the ORC unit **203**. In other words, the condenser or heat sink **211** may cool the second working fluid or working fluid of the ORC unit **203** causing the second working fluid or working fluid of the ORC unit **203** to condense or change phases from vapor to liquid. The ORC unit **203** may also include a working fluid reservoir **298** to store an amount of working fluid, e.g., in a liquid state, to ensure continuous operation of the ORC unit **203**. The liquid state working fluid, whether from the working fluid reservoir **298** or directly from the condenser/heat sink **211**, may be pumped, via pump **244**, back to the heat exchanger **205**. Further, the pressure prior to and after pumping, e.g., as measured by the pressure sensors **238**, **246**, may be monitored to ensure that the working fluid remains at a ORC unit or working fluid loop pressure rating.

(35) As illustrated in FIG. 2C, the well may include an intermediary heat exchanger **219**. In such examples, an ORC unit **203** may not be configured for high-pressure heat exchange. As such, an intermediary heat exchanger **219** may be disposed nearby the high-pressure heat exchanger **250**, nearby the ORC unit **203**, or disposed at some other point in between to alleviate such issues. The intermediary heat exchanger **219** may include a working fluid, also referred to as an intermediary working fluid, to flow through the intermediary loop **223**. The intermediary fluid may include water, a water and glycol mixture, or other organic fluid exhibiting a higher vaporous phase change threshold than the working fluid of the ORC unit **203**. In an embodiment, the intermediary heat exchanger **219** may include sensors, meters, transducers and/or other devices at various positions throughout to determine characteristic of fluids flowing therein, similar to that of the high-pressure heat exchanger **250**.

(36) As illustrated in FIG. 2D, rather than utilizing an ORC unit **203** that may not withstand high pressure, a high-pressure ORC unit or an ORC unit with integrated high-pressure heat exchanger **250** may be utilized for geothermal power generation. In such examples, the components, equipment, and devices may be similar to those described above. In another example, such a system, as illustrated in FIG. 2D, may include a heat sink **236** utilizing a cooled flow of wellhead fluid to cool the flow of working fluid. In such examples, as the flow of wellhead fluid passes

through a choke valve **252**, the pressure of the flow of wellhead fluid may be reduced, e.g., for example, from about 15,000 PSI to about 1,500 PSI, from about 15,000 PSI to about 200 PSI, from about 15,000 PSI to about 100 PSI, about 15,000 PSI to about 50 PSI or lower, from about 10,000 PSI to about 200 PSI or lower, from about 5,000 PSI to about 200 PSI or lower, or from 15,000 PSI to lower than 200 PSI. In such examples, the temperature from a point prior to the choke valve **252** and after the choke valve **252**, e.g., a temperature differential, may be about 100 degrees Celsius, about 75 degrees Celsius, about 50 degrees Celsius, about 40 degrees Celsius, about 30 degrees Celsius, and lower. For example, the temperature of the wellhead fluid prior to the choke valve **252** may be about 50 degrees and higher, while, after passing through the choke valve **252**, the temperature, as measured by the temperature sensor **259**, may be about 30 degrees Celsius, about 25 degrees Celsius, about 20 degrees Celsius, to about 0 degrees Celsius.

(37) The system may include, as noted, a temperature sensor **259** and pressure sensor **257** to determine the temperature of the wellhead fluid after the choke valve **252**. The system may include temperature sensor **240** to determine the temperature of the working fluid or ORC fluid exiting the heat sink **236** and temperature sensor **238** to determine the temperature of the working fluid or ORC fluid entering the heat sink **236**. The pressure and/or temperature of the wellhead fluid may be used to determine whether the heat sink **236** may be utilized based on pressure rating of the heat sink **236** and/or a liquid phase change threshold of the working fluid. In other words, if the flow of wellhead fluid is at a temperature sufficient to cool the working fluid and/or below a pressure rating of the heat sink **236**, the heat sink valve **254** may open to allow wellhead fluid to flow through the heat sink **236** to facilitate cooling of the working fluid. In another embodiment, the heat sink valve **254** may initially be fully or partially open. The temperature of the working fluid or ORC fluid may be measured as the fluid enters the heat sink **236** and exits the heat sink **236**. If the temperature differential indicates that there is no change or an increase in temperature, based on the temperature of the working fluid or ORC fluid entering the heat sink **236** and then leaving the heat sink **236**, then the heat sink valve **254** may be closed. Temperature sensors **238**, **240**, **256**, **262**, and pressure sensors **258**, **260** may be disposed within the heat sink **236** to ensure that the temperature of the wellhead fluid is suitable for cooling the working fluid and that the pressure of wellhead fluid does not exceed the pressure rating of the heat sink **236**.

(38) FIG. 2D also illustrates two flow meters **277**, **279** disposed prior to the first wellhead fluid valve **210** and first heat exchanger valve **208**. Such flow meters may measure the flow of the wellhead fluid at the point where the meter is disposed. Utilizing flow measurements may allow for fine-tuning or adjustment of the open percentage or position of the valves included in the system. Such fine-tuning or adjustment may ensure that the production of hydrocarbons at the well is not impeded by the use of the high-pressure heat exchanger. Other flow meters may be disposed at various other points of the system, e.g., after the first wellhead fluid valve **210**, prior to or after the choke valve **252**, at a point after the heat sink **236**, and/or at various other points in the system. As stated, these flow meter may be utilized to ensure proper flow wellhead fluid throughout the system. In an example, the sensors and/or meters disposed throughout the system may be temperature sensors, densitometers, density measuring sensors, pressure transducers, pressure sensors, flow meters, mass flow meters, Coriolis meters, spectrometer, other measurement sensors to determine a temperature, pressure, flow, composition, density, or other variable as will be understood by those skilled in the art, or some combination thereof. Further, the sensors and/or meters may be in fluid communication with a liquid to measure the temperature, pressure, or flow or may indirectly measure flow (e.g., an ultrasonic sensor). In other words, the sensors or meters may be a clamp-on device to measure flow indirectly (such as via ultrasound passed through the pipe to the liquid).

(39) As illustrated in FIG. 2E, a controller **272** may be included at the well. The controller **272** may be utilized in any of the previous or following drawings. The controller **272** may include one or more controllers, a supervisory controller, and/or a master controller. The controller **272** may

connect to all the equipment and devices shown, including additional equipment and devices not shown, and may transmit control signals, receive or request data or measurements, control pumps, monitor electricity generated, among other things (see **274**). The controller **272** may, in another example, control a subset of the components shown. In another example, a controller may be included in an ORC unit (see **203** in FIGS. 2A through 2C). The controller **272** may connect to and control the controller in the ORC unit **203**. The controller **272** may transmit signals to the various control valves to open and close the valves by determined amounts or percentages or fully open or close the valves. The controller **272** may further determine, via a meter or other device or sensor, an amount of electrical power generated or being generated by the generator **234** or ORC unit **203**.

(40) As illustrated in FIG. 2F, the system may include another heat sink **241**, in the case that the cooling offered by the flow of wellhead fluid in heat sink **236** is not sufficient. In an example, the heat sink **241** may be a fin fan cooler, a heat exchanger, a condenser, any other type of heat sink, a sing-pass parallel flow heat exchanger, a 2-pass crossflow heat exchanger, a 2-pass countercurrent heat exchanger, or other type of apparatus. As illustrated in FIG. 2G, the system may include a regenerator **290**. In such examples, the working fluid may flow through a first fluid path of the regenerator **290**. After the working fluid is cooled by a primary heat sink (e.g., heat sink **236**), the working fluid may flow back through another fluid path of the regenerator **290**. As such, the heat from the first fluid path may pre-heat the working fluid, while the second fluid path may offer some level of cooling to the working fluid.

(41) As illustrated in FIG. 2H, the system may include a gas expander **291**. In an example, the gas expander **291** may be a turbine expander, positive displacement expander, scroll expander, screw expander, twin-screw expander, vane expander, piston expander, other volumetric expander, and/or any other expander suitable for an ORC operation or cycle. For example and as illustrated, the gas expander **291** may be a turbine expander. As gas flows through the turbine expander, a rotor **293** connected to the turbine expander may begin to turn, spin, or rotate. The rotor **293** may include an end with windings. The end with windings may correspond to a stator **294** including windings and a magnetic field. As the rotor **293** spins within the stator **294**, electricity may be generated. Other generators may be utilized, as will be understood by those skilled in the art. The generator **293** may produce DC power, AC power, single phase power, or three phase power.

(42) FIG. 3A and FIG. 3B are block diagrams illustrating other novel implementations of a geothermal power generation enabled well to provide electrical power to one or more of in-field equipment, equipment at other wells, energy storage devices, and the grid power structure, according to one or more embodiment of the disclosure. FIGS. 3A and 3B may represent a side-view perspective block diagram of a well and the components at the well. In an example, wellhead fluid may flow from underground **304**. The wellhead fluid flow **308** may include hydrocarbons or a mixture of hydrocarbons and other fluids, e.g., water, chemicals, fluids leftover from fracturing operations, other residuals, and/or other fluids as will be understood by those skilled in the art. The wellhead fluid may flow from a wellbore.

(43) As the wellhead fluid flows from the wellhead **306**, the wellhead fluid may flow to the high-pressure heat exchanger, through a bypass pipe, and/or a combination thereof based on various factors or characteristics, e.g., wellhead fluid temperature and/or pressure and/or working fluid temperature. For example, if the wellhead fluid flow **308** is above a vaporous phase change threshold for a working fluid flow **310**, then valve **332** may open, at least partially, to allow the wellhead fluid flow to the high pressure heat exchanger **312**. In such examples, the wellhead fluid may continue to flow through the primary or bypass wellhead fluid pipe. As such, valve **330** may remain open, whether completely or at a certain percentage. From the high-pressure heat exchanger **312**, the wellhead fluid may flow back to the primary or bypass wellhead fluid pipe, to a condenser **316** or other cooling apparatus, and/or a combination thereof. If the wellhead fluid is at a temperature to provide cooling to the working fluid flow **310**, then valve **336** may open to allow wellhead fluid to flow therethrough. In such examples, the valve **334** may close to prevent

wellhead fluid from flowing back. If the wellhead fluid is not at a temperature to allow for cooling of the working fluid flow **310**, then valve **336** may close or remained closed and valve **334** may open or remain open. From the condenser **316** or the primary or bypass wellhead fluid pipe, the wellhead fluid may flow to in-field equipment **316**, storage tanks, and/or other processing equipment at the well. The valves described above may be controlled via controller **320**.

(44) In another embodiment, rather than basing the opening and closing of valve **332** and/or valve **336** on wellhead fluid flow **308** temperature, the valve **332** and/or valve **336** may be opened or closed based on the temperature of the working fluid flow **310**. For example, prior to activating the wellhead **306** (e.g., allowing wellhead fluid to flow or pumping wellhead fluid from the wellhead **306**), valve **332** may be open, fully or partially. As the wellhead fluid flows through the high-pressure heat exchanger **312**, the temperature of the working fluid flow **310** may be measured. Based on the working fluid flow **310** temperature, taken at continuously or at periodic intervals, and after a specified period of time, if the working fluid flow **310** does not reach a vaporous phase change temperature, then valve **332** may be closed. Further, such operations may be performed in conjunction with measuring wellhead fluid flow **308** and opening or closing valve **332** based on such measurements.

(45) The wellhead fluid flowing through the high-pressure heat exchanger **312** may be at a temperature to facilitate heat transfer to a working fluid flow **310**. The working fluid may further flow, as a vaporous state working fluid flow to an ORC expander/generator **314**. The vaporous state working fluid may cause the ORC expander/generator **314** to generate electrical power to be utilized at equipment at the well (e.g., in-field equipment), energy storage device, or a grid power structure (via a transformer and power lines). The working fluid may then flow to a condenser **316** or other cooling apparatus. The condenser **316** or other cooling apparatus may facilitate cooling of the working fluid flow **310** via the wellhead fluid flow, air, another liquid, and/or other types of heat sinks or heat exchangers. The liquid state working fluid may then flow back to the high-pressure heat exchanger **312**.

(46) In another embodiment, the high-pressure heat exchanger **312** may connect to an ORC unit/module **340** or one or more ORC units or modules. The number of ORC units/modules may scale based on power to be utilized by in-field equipment, the amount or potential capacity of electricity generation at the well, and/or other factors. After production of hydrocarbons begins, additional ORC units/modules may be added at the well or existing ORC units/modules may be removed from the well.

(47) FIG. 4A and FIG. 4B are simplified diagrams illustrating a control system for managing the geothermal power production at a well, according to one or more embodiment of the disclosure. A master controller **402** may manage the operations of geothermal power generation at a wellhead during hydrocarbon production. The master controller **402** may be one or more controllers, a supervisory controller, programmable logic controller (PLC), a computing device (such as a laptop, desktop computing device, and/or a server), an edge server, a cloud based computing device, and/or other suitable devices. The master controller **402** may be located at or near the well. The master controller **402** may be located remote from the well. The master controller **402**, as noted, may be more than one controller. In such cases, the master controller **402** may be located near or at various wells and/or at other off-site locations. The master controller **402** may include a processor **404**, or one or more processors, and memory **406**. The memory **406** may include instructions. In an example, the memory **406** may be a non-transitory machine-readable storage medium. As used herein, a “non-transitory machine-readable storage medium” may be any electronic, magnetic, optical, or other physical storage apparatus to contain or store information such as executable instructions, data, and the like. For example, any machine-readable storage medium described herein may be any of random access memory (RAM), volatile memory, non-volatile memory, flash memory, a storage drive (e.g., hard drive), a solid state drive, any type of storage disc, and the like, or a combination thereof. As noted, the memory **406** may store or include instructions executable

by the processor **404**. As used herein, a “processor” may include, for example one processor or multiple processors included in a single device or distributed across multiple computing devices. The processor may be at least one of a central processing unit (CPU), a semiconductor-based microprocessor, a graphics processing unit (GPU), a field-programmable gate array (FPGA) to retrieve and execute instructions, a real time processor (RTP), other electronic circuitry suitable for the retrieval and execution instructions stored on a machine-readable storage medium, or a combination thereof.

(48) As used herein, “signal communication” refers to electric communication such as hard wiring two components together or wireless communication for remote monitoring and control/operation, as understood by those skilled in the art. For example, wireless communication may be Wi-Fi®, Bluetooth®, ZigBee, cellular wireless communication, satellite communication, or forms of near field communications. In addition, signal communication may include one or more intermediate controllers or relays disposed between elements that are in signal communication with one another.

(49) The master controller **402** may include instructions **408** to measure temperature at various points or locations of the system (e.g., as illustrated in, for example, FIG. 2E). For example, temperature may be measured at a wellhead fluid temperature sensor **1 416**, heat exchanger temperature sensors **418**, wellhead fluid temperature sensor **2 420**, condenser temperature sensors **422**, and/or at various other points in the system **400**. Other characteristics may be measured as well, such as flow, density, pressure, composition, or other characteristics related to the wellhead fluid and/or working fluid.

(50) Utilizing the characteristics noted above, the master controller **402** may control various aspects of the system **400**. For example, the master controller **402** may include flow control adjustment instructions **412**. The system **400** may include one or more valves placed in various locations (For example, but not limited to, FIGS. 2A through 2H). Valves of the system **400** may include a wellhead fluid valve **1 426**, heat exchanger valves **428**, wellhead fluid valve **2 430**, and condenser valves **432**. As noted other valves may be included in the system and controlled by the master controller **402**. The valves may operate to adjust flow based on a number of factors. Such factors may include temperature of a wellhead fluid, flow rate of the wellhead fluid, temperature of the working fluid, flow rate of the working fluid, pressure of the wellhead fluid at various points in the system, pressure of the working fluid at various points in the system, and/or some combination thereof.

(51) In an example, the system **400** may include a user interface **436**, e.g., such as a monitor, display, computing device, smartphones, tablets, and other similar devices as will be understood by those skilled in the art. A user may view data, enter thresholds or limits, monitor status of the equipment, and perform other various tasks in relation to the equipment at the well. For example, a specific flow rate may be set for hydrocarbon production. As a wellhead begins producing hydrocarbons (e.g., wellhead fluids begin flowing from a wellhead), the master controller **402** may monitor flow rate and compare the flow rate to the threshold either set by a user or pre-set in the master controller **402**. If the master controller **402** determines that the heat exchanger valves **428** (e.g., via flow control adjustment instructions **412**) should be open or are open and that the flow of wellhead fluid is higher or lower than the threshold, the master controller **402** may adjust the appropriate valves, e.g., wellhead fluid valve **426** and/or heat exchanger valves **428**. The valve associated with the primary or bypass wellhead fluid pipe, e.g., wellhead fluid valve **426**, may open or close by varying degrees based on such determinations.

(52) In another example, the master controller **402** may include instructions **410** to control a pump for the ORC unit, e.g., working fluid pump **424**. If heat exchanger valves **428** and condenser valves **432** are closed, the master controller **402** may transmit a signal to shut down or cease operation of the working fluid pump **424**, if the working fluid pump **424** is operating. The master controller **402** may further transmit a signal, based on the heat exchanger valves **428** being open, to initiate or start operations of the working fluid pump **424**. The working fluid pump **424** may be a fixed pressure

pump or a variable frequency pump. The master controller **402** may further include instructions **414** to monitor the power output from an ORC unit or from expanders/generators **434** (e.g., expander/generator A **434A**, expander/generator B **434B**, and/or up to expander/generator N **434N**). If the system **400** utilizes ORC units, the master controller **402** may determine the electrical power generated or output based on an output from, for example, ORC unit controller A **438A**, ORC unit controller B **438B**, and/or up to ORC unit controller N **438N**. If the power output drops to an uneconomical or unsustainable level or electrical power generation ceases completely while the heat exchanger valves **428** are open, the master controller **402** may transmit signals to close the heat exchanger valves **428**. In another example, the master controller **402** may monitor electrical power output from other wells. The master controller **402** may monitor or meter the amount of electrical power being utilized at each of the wells and/or the amount of electrical power being generated at each of the wells. If an excess of electrical power exists, the master controller **402** may transmit signals causing the excess energy at any particular well to be stored in energy storage devices, transmitted to the grid, and/or transmitted to another well. If a deficit of electrical power exists, the master controller **402** may transmit a signal causing other wells to transmit electrical power to the well experiencing an electrical power deficit. In another example, the metered electrical power may be utilized for commercial trade, to determine a cost of the electricity generated, and/or for use in determining emissions or emission reductions through use of an alternate energy source (e.g., geothermal power).

(53) In another example, the master controller **402** may include instructions to maximize energy output from an ORC unit. In such examples, the ORC unit may be connected to a plurality of high-pressure heat exchangers. Further, each of the high-pressure heat exchangers may connect to one or more wellheads. As a wellhead produces a wellhead fluid, the pressure and temperature of the wellhead fluid may vary, over time, as well as based on the location of the wellhead. The master controller **402** may determine the temperature of the wellhead fluid at each high-pressure heat exchanger and/or the temperature of the working fluid in each high-pressure heat exchanger. Based on these determinations, the master controller **402** may open/close valves associated with one or more particular high-pressure heat exchangers to ensure the most efficient heat transfer. Further, the master controller **402** may determine the amount of electrical power output from the ORC unit. Based on a power rating of the ORC unit (e.g., the maximum power output the ORC unit is able to produce) and/or the amount of electrical power output from the ORC unit, the master controller **402** may adjust valves associated with the one or more particular high-pressure heat exchangers to thereby increase electrical power output. Additional ORC units may be utilized and electrical power output for each may be optimized or efficiently generated. The master controller **402** may determine, for each ORC unit, the optimal amount or efficient amount of heated working fluid flowing from each high-pressure heat exchanger to ensure the highest amount of electrical power possible is generated per ORC unit or that each ORC unit meets a preselected electrical power output threshold. In such examples, each ORC unit may be connected to each high-pressure heat exchanger and the master controller **402** may determine which set of valves to open/close based on such an optimization or electrical power output threshold.

(54) In another example, the master controller **402** may include failover instructions or instructions to be executed to effectively reduce or prevent risk. The failover instructions may execute in the event of ORC unit and/or high-pressure heat exchanger failure or if an ORC unit and/or high-pressure heat exchanger experiences an issue requiring maintenance. For example, the ORC unit and/or high-pressure heat exchanger may have various sensors or meters. Such sensors or meters, when providing measurement to the master controller **402**, may indicate a failure in the ORC unit and/or high-pressure heat exchanger. In another example, the master controller **402** may include pre-determined parameters that indicate failures. If the master controller **402** receives such indications, the master controller **402** may open, if not already open, the wellhead fluid valve **1 426** and wellhead fluid valve **2 430**. After the wellhead fluid valve **1 426** and wellhead fluid valve **2 430**

are opened, the master controller **402** may close the heat exchanger valves **428**, the condenser valves **432**, or any other valve associated with the flow of fluid to the ORC unit and/or high-pressure heat exchanger. In such examples, the master controller **402** may prevent further use of the ORC unit and/or high-pressure heat exchanger until the issue or failure indicated is resolved. Such a resolution may be indicated by a user via the user interface **436** or based on measurements from sensors and/or meters.

(55) In another example, the master controller **402** may, as noted, determine an amount of electrical power output by an ORC unit. The master controller **402** may additionally determine different characteristics of the electrical power output. For example, the master controller **402** may monitor the output voltage and frequency. Further, the master controller **402** may include pre-set or predetermined thresholds, limits, or parameters in relation to the monitored characteristics of the electrical power output. Further still, the master controller **402** may connect to a breaker or switchgear. In the event that the master controller **402** detects that an ORC unit exceeds any of the thresholds, limits, and/or parameters, the master controller **402** may transmit a signal to the breaker or switchgear to break the circuit (e.g., the flow of electricity from the ORC unit to a source) and may shut down the ORC unit (e.g., closing valves preventing further flow to the ORC unit, as described above).

(56) FIG. 5 is a flow diagram of geothermal power generation in which, when a wellhead fluid is at or above a vaporous phase change temperature threshold, heat exchanger valves may be opened to allow wellhead fluid to flow therethrough, thereby facilitating heating of a working fluid for use in a ORC unit, according to one or more embodiment of the disclosure. The method is detailed with reference to the master controller **402** and system **400** of FIGS. 4A and 4B. Unless otherwise specified, the actions of method **500** may be completed within the master controller **402**.

Specifically, method **500** may be included in one or more programs, protocols, or instructions loaded into the memory of the master controller **402** and executed on the processor or one or more processors of the master controller **402**. The order in which the operations are described is not intended to be construed as a limitation, and any number of the described blocks may be combined in any order and/or in parallel to implement the methods.

(57) At block **502**, the master controller **402** may determine whether the wellhead is active. Such a determination may be made based on sensors located at or near the wellhead, e.g., a pressure sensor indicating a pressure or a flow meter indicating a flow of a wellhead fluid or hydrocarbon stream from the wellhead. In other examples, a user may indicate, via the user interface **436**, that the wellhead is active. If the wellhead is not active, the master controller **402** may wait for a specified period of time and make such a determination after the period of time. In an example, the master controller **402** may continuously check for wellhead activity.

(58) At block **504**, in response to wellhead activity or during hydrocarbon production, the master controller **402** may determine a wellhead fluid temperature. The wellhead fluid temperature may be measured by a wellhead fluid temperature sensor **1 416** disposed at or near the wellhead. The wellhead fluid temperature sensor **1 416** may be disposed on or in a pipe. In an example, various other temperature sensors may be disposed at other points in the system **400**, e.g., heat exchanger temperature sensor **418**, wellhead fluid temperature sensor **2 420**, condenser temperature sensor **422**, and/or other temperature sensors. The temperature measurements provided by such sensors may be utilized by the master controller **402** to determine which valves to open or close.

(59) At block **506**, the master controller **402** may determine whether the wellhead fluid is at or above a vaporous phase change temperature threshold of a working fluid. In such examples, the vaporous phase change temperature may be based on the working fluid of the ORC unit. For example, for pentafluoropropane the vaporous phase change temperature or boiling point may be 15.14 degrees Celsius. In another example, or factors may be taken into account when determining whether to open heat exchanger valves **428**. For example, whether the pressure is within operating range of a high-pressure heat exchanger, whether the flow rate at a primary or bypass pipeline is

sufficient to prevent impedance of hydrocarbon production, whether power generation costs are offset by power generation needs, among other factors.

(60) At block **508**, if the wellhead fluid is at or above the vaporous phase change temperature, the master controller **402** may transmit a signal to heat exchanger valves **428** to open to a specified degree. In an example, the heat exchanger valves **428** may be may be fully opened or partially opened. The degree to which the heat exchanger valves **428** opens may depend on the temperature of the wellhead fluid, the flow rate of the wellhead fluid, and/or the pressure of the wellhead fluid.

(61) At block **510**, the master controller **402** may close wellhead fluid valves (e.g., wellhead fluid valve **1 426**) to divert a portion of the flow of wellhead fluids to the high-pressure heat exchanger. The wellhead fluid valves (e.g., wellhead fluid valve **1 426**) may close partially or completely, depending on various factors, such as heat exchanger flow capacity, current flow rate, current pressure, current temperature, among other factors. Once the wellhead fluid valves (e.g., wellhead fluid valve **1 426**) are closed, at block **512**, a working fluid pump **424** of the ORC unit may begin pumping the working fluid through the ORC loop. At block **514**, the master controller **402** may determine whether electricity is being generated. If not, the master controller **402** may check if the wellhead is still active and, if the wellhead is still active, the master controller **402** may adjust the valves (e.g., wellhead fluid valve **1 426** and heat exchanger valves **428**) as appropriate (e.g., increasing flow through the heat exchanger to facilitate an increase in heat transfer).

(62) At block **516**, if the wellhead fluid is lower than the vaporous phase change temperature, the master controller **402** may open or check if the wellhead fluid valves (e.g., wellhead fluid valve **1 426**) are open. Further, the wellhead fluid valves may already be open to a degree and, at block **516**, may open further or fully open, depending on desired wellhead fluid flow. In an example, the wellhead fluid valve (e.g., wellhead fluid valve **1 426**) may be used, with or without a separate choke valve, to choke or partially choke the wellhead fluid flow. Further, once the wellhead fluid valves are open, at block **518**, the master controller **402** may close the heat exchanger valves **428** fully or partially in some cases.

(63) FIG. **6** is another flow diagram of geothermal power generation in which, when a wellhead fluid is at or above a vaporous phase change temperature, heat exchanger valves may be opened to allow wellhead fluid to flow therethrough, thereby facilitating heating of a working fluid for use in a ORC unit, according to one or more embodiment of the disclosure. The method is detailed with reference to the master controller **402** and system **400** of FIGS. **4A** and **4B**. Unless otherwise specified, the actions of method **600** may be completed within the master controller **402**.

Specifically, method **600** may be included in one or more programs, protocols, or instructions loaded into the memory of the master controller **402** and executed on the processor or one or more processors of the master controller **402**. The order in which the operations are described is not intended to be construed as a limitation, and any number of the described blocks may be combined in any order and/or in parallel to implement the methods.

(64) Blocks **602** through **610** correspond to blocks **502** through **610**, as described above. Once it has been determined that the heat exchanger valves **428** should be open and after the heat exchanger valves **428** open, wellhead fluid may flow through the heat exchanger and/or the primary or bypass wellhead fluid pipe to a choke valve. The choke valve may reduce the pressure of the wellhead fluid and, thus, reduce the temperature of the wellhead fluid. At block **612**, the master controller **402** may determine the reduced pressure wellhead fluid temperature at or near a condenser or heat sink valve based on a measurement from the condenser temperature sensors **422**. The master controller **402** may, at block **614**, determine whether the wellhead fluid is at cool enough temperatures to facilitate cooling of a working fluid flow. The working fluid may have a condensation point or a temperature at which the working fluid changes phase from a vapor to a liquid. Such a temperature may be utilized as the threshold for such determinations.

(65) At block **616**, if the temperature is cool enough, the master controller **402** may open the condenser valves **432**, allowing wellhead fluid to flow through the condenser or other cooling

apparatus. At block **617** the master controller **402** may transmit a signal to the working fluid pump **424** to start or begin pumping working fluid through an ORC loop. In another example, at block **619**, if the temperature of the working fluid is not cool enough to facilitate cooling of the working fluid to any degree, the master controller **402** may close the condenser valves **432**. In another example, the master controller **402** may determine the temperature of the reduced pressure wellhead fluid flow and whether the temperature of the reduced pressure wellhead fluid flow, in conjunction with a primary or secondary cooler, may cool the working fluid to a point. The master controller **402**, in such examples, may consider the temperature of the working fluid entering the condenser and the temperature of the reduced pressure wellhead fluid flow at or near the condenser valve **432**.

(66) As noted and described above, the master controller **402** may, at block **618**, determine whether electric power is generated. In another example, if the wellhead temperature is not high enough to produce geothermal power, the master controller **402** may, at block **620**, open the wellhead fluid valves. At block **622**, the master controller **402** may close, if the heat exchanger valves **428** are open, the heat exchanger valves **428**. Finally, at block **624**, the master controller **402** may close condenser valves **432**.

(67) FIG. 7A is a flow diagram of geothermal power generation in which, when a working fluid or ORC fluid is at or above a vaporous phase change temperature threshold, heat exchanger valves may remain open to allow wellhead fluid to flow therethrough, thereby facilitating heating of the working fluid or ORC fluid for use in an ORC unit, according to one or more embodiment of the disclosure. The method is detailed with reference to the master controller **402** and system **400** of FIGS. 4A and 4B. Unless otherwise specified, the actions of method **700** may be completed within the master controller **402**. Specifically, method **700** may be included in one or more programs, protocols, or instructions loaded into the memory of the master controller **402** and executed on the processor or one or more processors of the master controller **402**. The order in which the operations are described is not intended to be construed as a limitation, and any number of the described blocks may be combined in any order and/or in parallel to implement the methods.

(68) At block **702**, the master controller **702** may determine whether the wellhead is active. Such a determination may be made based on sensors located at or near the wellhead, e.g., a pressure sensor indicating a pressure or a flow meter indicating a flow of a wellhead fluid or hydrocarbon stream from the wellhead. In other examples, a user may indicate, via the user interface **436**, that the wellhead is active. If the wellhead is not active, the master controller **402** may wait for a specified period of time and make such a determination after the specified period of time. In an example, the master controller **402** may continuously check for wellhead activity.

(69) At block **704**, in response to wellhead activity or during hydrocarbon production, the master controller **402** may open heat exchanger valves **428**. At block **706**, the master controller **402** may close wellhead fluid valve **1 426**, at least partially. At block **708**, when the heat exchanger valves **428** is open and wellhead fluid valve **1 426** is fully or partially closed, working fluid or ORC fluid may be pumped through the ORC unit.

(70) At block **710**, the master controller **402** may measure the temperature of the working fluid or ORC fluid. At block **712**, the master controller **402** may determine whether the wellhead fluid is at or above a vaporous phase change temperature threshold. In such examples, the vaporous phase change may include when the working fluid or ORC fluid changes from a liquid to a vapor or gas. For example, for pentafluoropropane the vaporous phase change temperature or boiling point may be 15.14 degrees Celsius. In another example, other factors may be taken into account when determining whether to maintain an open percentage of the heat exchanger valves **428**. For example, whether the pressure is within operating range of a high-pressure heat exchanger, whether the flow rate at a primary or bypass pipeline is sufficient to prevent impedance of hydrocarbon production, whether power generation costs are offset by power generation needs, among other factors.

(71) At block **712**, if the working fluid or ORC fluid is at or above the vaporous phase change temperature, the master controller **402** may determine whether electricity is generated at the ORC unit. If electricity is not generated, the master controller **402** may check, at block **702**, whether the wellhead is active and perform the operations of method **700** again.

(72) If the working fluid or ORC fluid, at block **712** is not at a vaporous phase change temperature, then, at block **714**, the master controller **402** may first determine whether a first specified period of time has lapsed. The first period of time may be period of time of sufficient length to determine whether or not the working fluid or ORC fluid may reach a vaporous phase change state. Such a first specified period of time may be about an hour or more, two hours, three hours, four hours, or some other length of time during wellhead activity.

(73) If the first specified period of time has not lapsed, at block **716**, the master controller **402** may wait a second specified period of time before measuring the temperature of the working fluid or ORC fluid. The second specified period of time may be less than the first specified period of time.

(74) If the first specified period of time has lapsed, then the master controller **402** may have determined that, based on the temperature of the working fluid or ORC fluid, that the wellhead fluid may not reach temperatures sufficient to cause a vaporous phase change of the working fluid or ORC fluid. As such, at block **718**, the master controller may close the open wellhead fluid valves and, at block **720**, close the heat exchanger valves **428**.

(75) FIG. **7B** is another flow diagram of geothermal power generation in which, when a working fluid or ORC fluid is at or above a vaporous phase change temperature threshold, heat exchanger valves may remain open to allow wellhead fluid to flow therethrough, thereby facilitating heating of the working fluid or ORC fluid for use in an ORC unit, according to one or more embodiment of the disclosure. The method is detailed with reference to the master controller **402** and system **400** of FIGS. **4A** and **4B**. Unless otherwise specified, the actions of method **701** may be completed within the master controller **402**. Specifically, method **701** may be included in one or more programs, protocols, or instructions loaded into the memory of the master controller **402** and executed on the processor or one or more processors of the master controller **402**. The order in which the operations are described is not intended to be construed as a limitation, and any number of the described blocks may be combined in any order and/or in parallel to implement the methods.

(76) Blocks for FIG. **7B** may perform the same functions as described for blocks of FIG. **7A**. As such, those blocks include the same numbers, such as block **702** through block **722**. In addition to those operations, at block **724** the master controller **402** may check whether the heat exchanger valves may be open. If they are not the master controller **402** may open the heat exchanger valves and close, at least partially, the wellhead fluid valves. If the heat exchanger valves are open, the master controller **402** may not change or adjust any of the valves open percentage, but rather continue to or start pumping the working fluid or ORC unit through the ORC unit.

(77) In addition, after the master controller **402** closes the heat exchanger valves at block **720**, the master controller **402** may determine, at block **726**, the wellhead fluid temperature at the heat exchanger. At block **728**, the master controller may determine whether the wellhead fluid is at a vaporous phase change temperature of the working fluid or ORC fluid. If the wellhead fluid temperature is less than such a value, the master controller **402** may wait and measure the temperature again after a period of time. If the wellhead fluid temperature is greater than or equal to such a value, the master controller **402** may perform the operations of method **701** starting at block **724** again.

(78) FIG. **8** is a flow diagram of geothermal power generation in which a working fluid flow is determined based on a preselected electrical power output threshold, according to one or more embodiment of the disclosure. The method is detailed with reference to the master controller **402** and system **400** of FIGS. **4A** and **4B**. Unless otherwise specified, the actions of method **800** may be completed within the master controller **402**. Specifically, method **800** may be included in one or more programs, protocols, or instructions loaded into the memory of the master controller **402** and

executed on the processor or one or more processors of the master controller **402**. The order in which the operations are described is not intended to be construed as a limitation, and any number of the described blocks may be combined in any order and/or in parallel to implement the methods. (79) At block **802**, each of one or more heat exchangers may be connected to one or more wellhead fluid lines. Each of the one or more wellhead fluid lines may correspond to a wellhead. At block **804**, an ORC unit may be connected to the one or more heat exchangers. In another example, the system **400** may include one or more ORC units and each of the one or more ORC units may connect to one or more heat exchangers or two or more heat exchangers.

(80) At block **806**, heat exchanger valves positioned between the one or more heat exchangers and the one or more wellhead fluid lines may be opened. Once opened, the heat exchanger valves may allow for continuous diversion of the flow of wellhead fluid through the heat exchanger. The flow of wellhead fluid through the heat exchanger may facilitate transfer of heat from the flow of wellhead fluid to a flow of working fluid or intermediate working fluid.

(81) At block **808**, ORC unit valves may be opened. The ORC unit valves may initially be fully opened or partially opened. The ORC unit valves, when open, may allow for working fluid from each of the heat exchangers to flow into the ORC unit. Each working fluid flow may be combined and may pass through the ORC unit.

(82) At block **810**, the master controller **402** may determine one or more operating conditions of the ORC unit and/or the system **400**. The one or more operating conditions may include the flow rate and/or pressure of working fluid flowing through each of the one or more heat exchangers, the flow rate and/or pressure of wellhead fluid flowing through each of the one or more heat exchangers or at any other point downstream of the wellhead, the temperature of the working fluid in each of the one or more heat exchangers, the temperature of wellhead fluid at each of the one or more heat exchangers, the temperature of the combined working fluid flow at the ORC unit, the electrical power output from the ORC unit, and/or the open position of each of the valves included in the system **400**.

(83) At block **812**, based on the determined operating conditions, the master controller **402** may determine an optimal or efficient working fluid flow of the ORC unit. The optimal or efficient working fluid flow may depend on the temperature of the combined working fluid flowing to the ORC unit. The other operating conditions, described above, may be utilized to determine the optimal or efficient working fluid flow. The optimal or efficient working fluid flow may comprise the combined flow of working fluid flowing into the ORC unit to thereby produce a maximum amount of electrical power possible or to enable the ORC unit to meet a preselected electrical power output threshold. The optimal or efficient working fluid flow may be at a temperature sufficient to produce such an amount of electrical power (e.g., a temperature greater than or equal to the boiling point of the working fluid within the ORC unit). The optimal or efficient working fluid flow may be indicated by the master controller **402** as one or more open positions for each of the ORC unit valves and/or heat exchanger valves.

(84) At block **814**, the master controller **402** may, based on a determined optimal or efficient working fluid flow, determine whether to adjust the one or more ORC unit valves. Other valves within the system **400** may be adjusted based on the optimal or efficient working fluid flow, such as one or more heat exchanger valves and/or one or more wellhead fluid valves. If it is determined that the ORC unit valves or any other valves are to be adjusted, the master controller **402**, at block **816**, may transmit a signal to the valve to be adjusted indicating a new open position or closed position for the valve to adjust to. After the signal is transmitted, the valve may automatically adjust to the position indicated. If the valve is not to be adjusted or after the valves have been adjusted, the master controller **402** may determine operating conditions again. In an example, the master controller **402** may wait for a period of time, allowing the system to adjust to the new temperatures and flow rates or to reach equilibrium, prior to determining the operating conditions.

(85) FIG. 9A, FIG. 9B, FIG. 9C, FIG. 9D, and FIG. 9E are block diagrams illustrating novel

implementations of one or more geothermal power generation enabled wells to provide electrical power to one or more of in-field equipment, equipment at one of the other wells, energy storage devices, and the grid power structure, according to one or more embodiment of the disclosure. As illustrated in FIG. 9A, various wells **902A**, **902B**, up to **902N** may be positioned in proximity to one another. For example, one well (e.g., well A **902A**) may be located at a distance of over about 1 mile, about 5 miles, about 10 miles, about 25 miles, or more to another well (e.g., well B **902B** or well N **902N**). Each well **902A**, **902B**, **902N** may include various in-field equipment **910A**, **910B**, **910N**. Each well **902A**, **902B**, **902N** may include one or more wellheads **904A**, **904B**, **904N**. Each wellhead **904A**, **904B**, **904N** may connect to a high-pressure heat exchanger **906A**, **906B**, **906N** and the high-pressure heat exchanger **906A**, **906B**, **906N** may connect to ORC equipment **908A**, **908B**, **908N**. The ORC equipment **908A**, **908B**, **908N** may generate electrical power. The electrical power may be provided to the in-field equipment **910A**, **910B**, **910N**. A surplus of electrical power may be provided to an energy storage device **920A**, **920B**, **920N**. The ORC equipment **908A**, **908B**, **908N** may also transmit electrical power energy to other wells, energy storage devices, or to a grid power structure **912**.

(86) As noted and as illustrated in FIG. 9B, the high-pressure heat exchanger N **906N** may connect to one or more wellheads **904A**, **904B**, **904N**. For example, a well N **902** may include 1, 2, 3, or more wellheads (e.g., wellhead A **904A**, wellhead B **904B**, and/or up to wellhead N **904N**). All of the wellheads **904A**, **904B**, **904N** may connect to one high pressure heat exchanger N **906N**. In another example, a well N **902N** may include one or more high-pressure heat exchangers, as illustrated in FIG. 9D and FIG. 9E. In such examples, the wellheads **904A**, **904B**, **904N** may connect to one or more of the high-pressure heat exchangers.

(87) As described and as illustrated in FIG. 9C, the well N **902N** may include one or more wellheads (e.g., wellhead A **904A**, wellhead B **904B**, and/or up to wellhead N **904N**), one or more sets of ORC equipment (e.g., ORC equipment A **908A**, ORC equipment B **908B**, and/or up to ORC equipment N **908N**), and one or more sets of in-field equipment (e.g., in-field equipment A **910A**, in-field equipment B **910B**, and/or up to in-field equipment N **910N**). As noted, a well N **902N** may include one high-pressure heat exchanger, as illustrated in FIG. 9D and FIG. 9E. The well **902** may further include additional high-pressure heat exchangers **906**.

(88) As illustrated in FIGS. 9D and 9E, a well N **902N** may include one or more wellheads (e.g., wellhead A **904A**, wellhead B **904B**, and/or up to wellhead N **904N**). Each of the one or more wellheads **904A**, **904B**, **904N** may correspond to a high-pressure heat exchanger (e.g., heat exchanger A **906A**, heat exchanger B **906B**, and/or up to heat exchanger N **906N**). In other examples, two or more high-pressure heat exchangers may correspond to a particular wellhead, while in other examples, two or more wellheads may correspond to a high-pressure heat exchanger. Further, a well N **902N** may include one ORC equipment A **908A** or unit, as illustrated in FIG. 9D. A well N **902N** may include two ORC equipment (e.g., ORC equipment A **908A** and ORC equipment B **908B**) or units or, in some cases, more. In such examples, each ORC equipment **908A**, **908B** may connect to each of the high-pressure heat exchangers **906A**, **906B**, **906N**. As wellhead fluid flows from a wellhead **904A**, **904B**, **904N**, the temperature and pressure may vary based on a number of factors (e.g., production, type of fluids, distance from the high-pressure heat exchanger, among other factors).

(89) As such, a working fluid of a particular high-pressure heat exchanger **906A**, **906B**, **906N** may be heated to a degree sufficient, insufficient, or more than sufficient to cause the working fluid of the ORC equipment **908A**, **908B** to exhibit a vaporous phase change. Since the temperature of the wellhead fluid varies, a controller (e.g., controller **916A**, **916B**, up to **916N** or master controller **918**, as illustrated in FIG. 9F) may determine the temperature of working fluid at each high-pressure heat exchanger **906** and determine the most efficient and/or optimal combination to be utilized for generating the most electrical power or for generating an amount of electrical power to meet a preselected electrical power output threshold at the ORC equipment **908A**, **908B**. The

electrical power output by the ORC equipment **908A**, **908B** may be measured by, for example, an electrical power meter or other device suitable to determine electrical power output. The controller may also determine the temperature of the combination of working fluid or intermediate working fluid entering the ORC equipment **908A**, **908B**, via, for example, one or more temperature sensors positioned at or near an inlet of the ORC equipment **908A**, **908B**. The inlet may allow working fluid or intermediate working fluid to flow into the ORC equipment **908A**, **908B**. The controller may determine the most efficient and/or optimal combination or amount of working fluid or intermediate working fluid from the one or more high-pressure heat exchangers **906A**, **906B**, **906N** based on a variety of factors noted above. For example, the most efficient and/or optimal combination or amount may be based on wellhead fluid temperature from each of the one or more wellheads **904A**, **904B**, **904N**, the working fluid or intermediate working fluid temperature at each of the high-pressure heat exchangers **906A**, **906B**, **906N**, and/or electrical power output (e.g., a measurement indicative of the electrical power generated) by the ORC equipment **908A**, **908B**. Other factors may include flow rate and pressure of each of the high-pressure heat exchangers **906A**, **906B**, **906N**, current open positions high-pressure heat exchanger valves, and/or current open positions of other valves included at the well N **902N**.

(90) For example, if high-pressure heat exchanger A **906A** includes a working fluid at a temperature slightly less than a temperature to cause vaporous phase change, then valves providing working fluid or intermediate working fluid from the high-pressure heat exchanger A **906A** to the ORC equipment **908A**, **908B** may be closed. In another example, if high-pressure heat exchanger B **906B** is providing working fluid at a temperature well above a temperature to cause vaporous phase change, then valves providing working fluid or intermediate working fluid from high-pressure heat exchanger B **906B** to ORC equipment A **908A** and/or to ORC equipment B **908B** may be adjusted to positions such that a greater portion of the working fluid or intermediate working fluid is transported to ORC equipment A **908A** and/or to ORC equipment B **908B**.

(91) In yet another example, all valves for allowing flow of working fluid or intermediate working fluid to the ORC equipment A **908A** and/or ORC equipment B **908B** may be, at least, in a partially open position. The temperature of the wellhead fluid and/or working fluid of each heat exchanger **906A**, **906B**, **906N** may be determined or measured. Further, the electrical power output of the ORC equipment A **908A** and/or ORC equipment B **908B** may be determined. The positions of each valve for allowing flow of working fluid or intermediate working fluid to the ORC equipment A **908A** and/or ORC equipment B **908B** may be adjusted to different partially open positions, fully opened positions, or fully closed positions. Such valve adjustments may be based on maximization of the resultant temperature and heat delivered to the ORC equipment A **908A** and/or ORC equipment B **908B** once the combined working fluid flows into the ORC equipment **908A** and/or ORC equipment B **908B**. The valve adjustments may be based on, rather than or in addition to other factors, the maximization of the electrical power output from the ORC equipment A **908A** and/or ORC equipment B **908B**. Valve adjustments may further be based on wellhead fluid temperature and/or some combination of the factors described herein.

(92) As noted and described above and as illustrated in FIG. **9F**, a controller (e.g., controller A **916A**, controller B **916B**, and/or up to controller N **916N**) may be included at one or more of the wells **902A**, **902B**, **902N**. The controller **916A**, **916B**, **916N** may control and monitor various aspects of the well **902A**, **902B**, **902N**. The controller **916A**, **916B**, **916N** of each well **902A**, **902B**, **902N** may connect to a master controller **918**, the master controller **918** may control the operations of the ORC equipment **908A**, **908B**, **908N**, as well as other equipment (e.g., valves and/or pumps) at each of the wells **902A**, **902B**, **902N**. As described above and illustrated in FIG. **9G**, the master controller **918** may connect to one or more wellheads **904A**, **904B**, **904N**. The ORC enabled wells **902A**, **902B**, **902N** may provide power to one or more of in-field equipment at the ORC enabled well **902A**, **902B**, **902N**, to an energy storage device, and/or a grid structure device (see **912**).

(93) This application is a continuation of U.S. application Ser. No. 17/305,297, filed Jul. 2, 2021,

titled “SYSTEMS FOR GENERATING GEOTHERMAL POWER IN AN ORGANIC RANKINE CYCLE OPERATION DURING HYDROCARBON PRODUCTION BASED ON WORKING FLUID TEMPERATURE”, which claims priority to and the benefit of U.S. Provisional Application No. 63/200,908, filed Apr. 2, 2021, titled “SYSTEMS AND METHODS FOR GENERATING GEOTHERMAL POWER DURING HYDROCARBON PRODUCTION,” the disclosures of which are incorporated herein by reference in their entirety.

(94) In the drawings and specification, several embodiments of systems and methods to provide geothermal power in the vicinity of a wellhead during hydrocarbon production have been disclosed, and although specific terms are employed, the terms are used in a descriptive sense only and not for purposes of limitation. Embodiments of systems and methods have been described in considerable detail with specific reference to the illustrated embodiments. However, it will be apparent that various modifications and changes can be made within the spirit and scope of the embodiments of systems and methods as described in the foregoing specification, and such modifications and changes are to be considered equivalents and part of this disclosure.

Claims

1. A system for generating geothermal power in the vicinity of a wellhead during hydrocarbon production, thereby to supply electrical power to one or more of in-field equipment, a grid power structure, or energy storage devices, the system comprising: a first temperature sensor to provide a first temperature, the first temperature defined by a temperature of a flow of working fluid; a heat exchanger valve positioned to divert flow of wellhead fluid from one or more wellheads; a controller configured to signal the heat exchanger valve to divert the flow of the wellhead fluid based on when the first temperature reaches a vaporous phase-change temperature threshold; a high-pressure heat exchanger including a first fluid path to accept and output the flow of wellhead fluid from the heat exchanger valve and a second fluid path to accept and output the flow of the working fluid, the high-pressure heat exchanger positioned to indirectly transfer heat from the flow of wellhead fluid to the flow of the working fluid, thereby to cause the working fluid to change phases from a liquid to a vapor; and a power generation unit including a generator, a gas expander, and a partial loop for the flow of the working fluid, the partial loop defined by a fluid path through a condenser, the gas expander, and a pump, the partial loop further defining a complete loop when connected to the second fluid path of the high-pressure heat exchanger, the flow of the working fluid, as a vapor, positioned to cause the generator to generate electrical power via rotation of the gas expander, the condenser positioned to cool the flow of the working fluid so that the cooled flow of the working fluid changes phase from the vapor to the liquid, the pump positioned to transport the liquid state working fluid from the condenser for heating.
2. The system of claim 1, further comprising a first wellhead fluid valve to adjust flow of wellhead fluid from the one or more wellheads based on the diversion of the flow of wellhead fluid to the heat exchanger valve.
3. The system of claim 1, further comprising another one or more power generation units connected to the high-pressure heat exchanger, and wherein each of the one or more power generation units and the another one or more power generation units includes a respective working fluid reservoir to store a corresponding working fluid.
4. The system of claim 1, wherein the first fluid path of the high-pressure heat exchanger is configured to withstand corrosion caused by the wellhead fluid via an anti-corrosive coating applied on the first fluid path or an injection point for anti-corrosive chemical additive injections into the first fluid path.
5. A system for generating power in thermal power operation in the vicinity of a wellhead during hydrocarbon production, thereby to supply electrical power to one or more of in-field equipment, a grid power structure, or energy storage devices, the system comprising: a first heat exchanger (a)

including a first fluid path to accept and output a flow of wellhead fluid and a second fluid path to accept and output a flow of a first working fluid and (b) positioned to indirectly transfer heat from the flow of wellhead fluid to the flow of the first working fluid; a first temperature sensor to provide a first temperature of the flow of the first working fluid output from the second fluid path; a first valve positioned to divert the flow of the wellhead fluid from one or more wellheads to the first fluid path; a controller configured to signal the first valve to divert the flow of the wellhead fluid based on (a) when the first temperature reaches a vaporous phase-change temperature threshold and (b) production of wellhead fluid from the one or more wellheads; a second valve to adjust the flow of wellhead fluid from the one or more wellheads downstream based on (a) diversion of the flow of wellhead fluid to the first valve, (b) when the first temperature reaches the vaporous phase-change temperature threshold, and (c) production of wellhead fluid from the one or more wellheads; and a thermal power generator including a second heat exchanger, the second heat exchanger (a) including a third fluid path to accept and output the flow of first working fluid from the first heat exchanger and a fourth fluid path to accept and output the flow of a second working fluid and (b) positioned to indirectly transfer heat from the flow of the first working fluid to the flow of the second working fluid and cause the second working fluid to change phases from a liquid to a vapor, thereby to cause the thermal power generator to generate electrical power.

6. The system of claim 5, further comprising: a second temperature sensor to provide to the controller a second temperature of the flow of the second working fluid output from the second heat exchanger, and wherein the controller is configured to signal the first valve to divert the flow of the wellhead fluid from the one or more wellheads to the first fluid path based on the second temperature.

7. The system of claim 5, further comprising: a third temperature sensor to provide to the controller a third temperature of the flow of the wellhead fluid accepted at the first heat exchanger, and wherein the controller is configured to signal the first valve to divert the flow of the wellhead fluid from the one or more wellheads to the first fluid path based on the third temperature.

8. The system of claim 5, further comprising a flow meter (a) positioned downstream of the first heat exchanger and the first valve and (b) configured to measure the flow of wellhead fluid downstream of the first heat exchanger and the first valve, and wherein the flow of wellhead fluid downstream of the first heat exchanger and the second valve indicates the production of wellhead fluid from the one or more wellheads.

9. A system for generating geothermal power in the vicinity of a wellhead during hydrocarbon production, thereby to supply electrical power to one or more of in-field equipment, a grid power structure, or energy storage devices, the system comprising: a first temperature sensor to provide a first temperature, the first temperature defined by a temperature of a flow of working fluid; a heat exchanger valve to divert flow of wellhead fluid from one or more wellheads based on when the first temperature reaches a vaporous phase-change temperature threshold within a selected period of time; a high-pressure heat exchanger including a first fluid path to accept and output the flow of wellhead fluid from the heat exchanger valve and a second fluid path to accept and output the flow of the working fluid, the high-pressure heat exchanger positioned to indirectly transfer heat from the flow of wellhead fluid to the flow of the working fluid, thereby to cause the working fluid to change phases from a liquid to a vapor; and a power generation unit including a generator, a gas expander, and a partial loop for the flow of the working fluid, the partial loop defined by a fluid path through a condenser, the gas expander, and a pump, the partial loop further defining a complete loop when connected to the second fluid path of the high-pressure heat exchanger, the flow of the working fluid, as a vapor, positioned to cause the generator to generate electrical power via rotation of the gas expander as defined by a power generation operation, the condenser positioned to cool the flow of the working fluid so that the cooled flow of the working fluid change phases from the vapor to the liquid, and the pump positioned to transport the liquid state working fluid from the condenser for heating.

10. A system for generating power in thermal power operation in the vicinity of a wellhead during hydrocarbon production, thereby to supply electrical power to one or more of in-field equipment, a grid power structure, or energy storage devices, the system comprising: a first heat exchanger (a) including a first fluid path to accept and output a flow of wellhead fluid and a second fluid path to accept and output a flow of a first working fluid and (b) positioned to indirectly transfer heat from the flow of wellhead fluid to the flow of the first working fluid; a first temperature sensor to provide a first temperature of the flow of the first working fluid output from the second fluid path; a first valve to divert the flow of the wellhead fluid from one or more wellheads to the first fluid path based on (a) when the first temperature reaches a vaporous phase-change temperature threshold and (b) production of wellhead fluid from the one or more wellheads; a second fluid valve to adjust the flow of wellhead fluid from the one or more wellheads downstream based on (a) diversion of the flow of wellhead fluid to the first valve, (b) when the first temperature reaches the vaporous phase-change temperature threshold, and (c) production of wellhead fluid from the one or more wellheads; a thermal power generator including a second heat exchanger, the second heat exchanger (a) including a third fluid path to accept and output the flow of first working fluid from the first heat exchanger and a fourth fluid path to accept and output the flow of a second working fluid and (b) positioned to indirectly transfer heat from the flow of the first working fluid to the flow of the second working fluid and cause the second working fluid to change phases from a liquid to a vapor so as to cause the thermal power generator to generate electrical power; and a second temperature sensor to provide a second temperature of the flow of the second working fluid output from the second heat exchanger, the first valve positioned to divert the flow of the wellhead fluid from one or more wellheads to the first fluid path based on the second temperature.

11. A system for generating power in thermal power operation in the vicinity of a wellhead during hydrocarbon production, thereby to supply electrical power to one or more of in-field equipment, a grid power structure, or energy storage devices, the system comprising: a high-pressure intermediate heat exchanger (a) including a first fluid path to accept and output a flow of wellhead fluid and a second fluid path to accept and output a flow of a first working fluid and (b) positioned to indirectly transfer heat from the flow of wellhead fluid to the flow of the first working fluid; a first temperature sensor to provide a first temperature of the flow of the first working fluid output from the second fluid path; a first valve to divert the flow of the wellhead fluid from one or more wellheads to the first fluid path based on (a) when the first temperature reaches a vaporous phase-change temperature threshold and (b) production of wellhead fluid from the one or more wellheads; a second valve to adjust the flow of wellhead fluid from the one or more wellheads downstream based on (a) diversion of the flow of wellhead fluid to the first valve, (b) when the first temperature reaches the vaporous phase-change temperature threshold, and (c) production of wellhead fluid from the one or more wellheads; a thermal power generator including a heat exchanger, the heat exchanger (a) including a third fluid path to accept and output the flow of first working fluid from the high-pressure intermediate heat exchanger and a fourth fluid path to accept and output the flow of a second working fluid and (b) to indirectly transfer heat from the flow of the first working fluid to the flow of the second working fluid and cause the second working fluid to change phases from a liquid to a vapor so as to cause the thermal power generator to generate electrical power; and a flow meter (a) positioned downstream of the high-pressure intermediate heat exchanger and the first valve and (b) configured to measure the flow of wellhead fluid downstream of the high-pressure intermediate heat exchanger and the first valve.

12. A system for generating geothermal power in the vicinity of a wellhead during hydrocarbon production, thereby to supply electrical power to one or more of in-field equipment, a grid power structure, or energy storage devices, the system comprising: a first temperature sensor to provide a first temperature, the first temperature defined by a temperature of a flow of working fluid; a heat exchanger valve to divert flow of wellhead fluid from one or more wellheads based on (a) when the first temperature reaches a vaporous phase-change temperature threshold and (b) a pressure of the

flow of wellhead fluid being below a pressure threshold; a high-pressure heat exchanger including a first fluid path to accept and output the flow of wellhead fluid from the heat exchanger valve and a second fluid path to accept and output the flow of the working fluid, the high-pressure heat exchanger positioned to indirectly transfer heat from the flow of wellhead fluid to the flow of the working fluid, thereby to cause the working fluid to change phases from a liquid to a vapor; and a power generation unit including a generator, a gas expander, and a partial loop for the flow of the working fluid, the partial loop defined by a fluid path through a condenser, the gas expander, and a pump, the partial loop further defining a complete loop when connected to the second fluid path of the high-pressure heat exchanger, the flow of the working fluid, as a vapor, positioned to cause the generator to generate electrical power via rotation of the gas expander as defined by a power generation operation, the condenser positioned to cool the flow of the working fluid so that the cooled flow of the working fluid changes phase from the vapor to the liquid, and the pump positioned to transport the liquid state working fluid from the condenser for heating.
